

Part III — The Desktop Deployment

ASSEMBLED SECTION DRAFT v1 (M0–M7, 2026-07-29) — section title provisional
pending arc decision

2026-07-29

1 From the Pilot to the Deployment

In the spring of 2022, the science library at the University of California, Irvine kept a shelf of Meta Quest 2 headsets on loan to the students of a biomedical-engineering elective.¹ Any student could borrow one at no cost and, through it, stand inside an operating room—a space few undergraduates will ever be admitted to. It is, we would later conclude, “indisputable that a majority of our students felt that VR increased their sense of presence and immersion” (“Phenomenological Evaluation” 390). Yet “[w]hile we had several Meta Quest 2’s housed in the school’s library, not a single student went there to check one out” (389). That untouched shelf is this section’s problem in miniature: a virtual world can offer presence and still lose its students to the more convenient copy of itself. What follows examines that problem at scale—three quarters of the same course, redeployed on ordinary computers to 241 students across the University of California—and asks what a virtual learning environment (VLE) must offer, beyond presence, to be worth entering.

Clinical immersion for non-medical staff within operating rooms and other restricted medical spaces is scarce by structure. The programs that admit undergraduates to such clinical settings draw substantially more applicants than they can serve, so that a majority of the students who seek a clinical-immersion experience will not have one—an inequity the biomedical-engineering literature documents against a clear demand.² The virtual clinical-immersion platform whose deployment this section reads was built against that scarcity: filmed operations, physician and engineer interviews, and explorable procedure rooms, delivered so that any enrolled student could be emplaced where only a limited few traditionally have access. In its first published test, a pilot cohort of twenty-two met the same operating-room footage in four competing formats: 2D video on YouTube, an interactive computer build,³ a Google Cardboard head mounted display (HMD),⁴ and a Meta Quest 2 virtual reality (VR) HMD. Interestingly, across twenty post-course responses we

¹Part III draws on several of my previous publications; the article referenced here—which is reproduced in Appendix 8—is the following, and will be cited parenthetically by abbreviated title and page number hereafter: King, Christine E., and Dalton Salvo. “Phenomenological Evaluation of an Undergraduate Clinical Needs Finding Skills Through a Virtual Reality Clinical Immersion Platform.” *Biomedical Engineering Education*, vol. 4, 2024, pp. 381–97, <https://doi.org/10.1007/s43683-024-00139-5>.

²Guilford, Kotche, and Schmedlen, surveying clinical-immersion experiences across biomedical-engineering programs, find that only 48% of applicants to clinical-immersion courses and 24% of applicants to immersion programs are able to participate annually (113–22). Moravec et al. report on one such program—a short-term clinical immersion for undergraduate biomedical engineering students—and the best practices such programs have developed under the same constraint of scarce placements and clear demand (217–23).

³The interactive computer build was developed in Unreal Engine 4 for standard PCs: essentially the same explorable virtual environment as the Meta Quest 2 build, but navigated by mouse and keyboard on a flat monitor, and delivering the traditional 2D videos rather than the stereoscopic footage (“Phenomenological Evaluation” 383).

⁴Google Cardboard is a minimal head-mounted display: a folded cardboard or plastic viewer holding two lenses, into which the user inserts a smartphone. The phone’s screen, split into a stereoscopic pair and viewed through the lenses, produces a three-dimensional image, and the phone’s own motion sensors track the rotation of the head. In this course the Cardboard route delivered the operating-room footage through YouTube’s VR mode—video only, with none of the interactive environment of the PC and Meta Quest 2 builds (“Phenomenological Evaluation” 383). YouTube has since discontinued this VR viewing mode, and the route the Cardboard format relied on is no longer available. Given the nature of this setup, of all VR methods it is the most prone to producing Alternate World Syndrome (AWS): motion sickness, physical disorientation, and physical discomfort. In Michael Heim’s definition, “AWS is the relativity sickness that comes from switching back-and-forth between the primary and virtual worlds” (*Virtual Realism* 182); the VR experience “can mix images and expectations from alternate worlds so as to distort our perception of the primary world, making us prone to errors in mismatched contexts” (185).

discovered a two-part verdict that would not line up. The highest level of presence⁵ went to the headset: 70% reported that the Meta Quest VR version elicited a greater felt sense of presence. Student preference, on the other hand, went the other way: 40% judged the traditional 2D YouTube videos most educationally beneficial. The two VR HMD formats together drew only that same share—20% choosing the Google Cardboard and 20% the Meta Quest 2—while 10% selected the PC build and 20% judged all versions equally beneficial (“Phenomenological Evaluation” 388). The students’ explanation as to why the 2D videos were perceived as being more educationally beneficial was largely grounded in physical discomfort⁶ and convenience.⁷ The loaner headsets went unused because, as one student described it, having to go to the library to check out the VR headset is “a hassle” (388).

In our previous article, we stated our expectations at the start. We did not expect virtual clinical immersion to rival physical presence in the operating room: “[i]t is unlikely that VR clinical immersion programs will be as pedagogically effective as in-person observation” (“Phenomenological Evaluation” 383). We hypothesized instead that the first-person perspective enabled by VR, “and the enhanced sense of presence and agency it generates, has the potential to be more beneficial to students’ development of such [clinical needs-finding] skills than 2D videos as it more closely simulates the experience of being physically present in the operating room” (383). The results substantiated half of that initial hypothesis—presence was at its highest in the Meta Quest VR environment, but student preference did not follow suit—and on the perceived pedagogical benefit, we were unable to render a verdict. Because the course’s commitment to open access let every student default to the most convenient format,⁸ the four versions could not compete on equal grounds: while a majority of students undeniably felt more present in virtual reality, we could not “conclusively claim that interactive VR learning environments are more or less pedagogically efficacious compared to 2D videos” (390).⁹ The pilot thus established two aspects and clearly drew a separation between them: presence was achieved to a much higher extent in VR, but was not enough to override the students’ preference for convenience when having to view the educational content at home on their own time. Whether the interactive virtual world was more pedagogically effective—whether felt presence converts into pedagogical worth—was left standing as a question our pilot’s own conditions could not answer.

Three concepts, built across the preceding parts, do the analytical work here. The first is the actualization chain: engagement understood as the actualization of desire, running from perception (A_1) through the charged image (*phantasma*, A_2) and committed judgment (*doxa*, A_3) into completed action (A_4)—a sequence a designed world can solicit, sustain, or fall silent at any station. The chain also has a zero point (A_0): before any of its stations can fire, the student and the world must reach one another at all—the threshold condition every later moment presupposes. The second is rhetorical incorporation: the union a virtual world achieves with its inhabitant, one in substance but dual in being—world-disclosedness, the world opened to a student

⁵We do not stipulate a formal definition of presence in the article; it uses the term in the sense standard to the VR-pedagogy literature it draws on—the felt sense of being physically present in the virtual rather than the actual environment—and treats it, alongside embodiment, as one of the “intensely subjective experiential phenomena” a phenomenological method is fitted to assess (“Phenomenological Evaluation” 382, 387). The construct is anchored to Makransky and Lilleholt (2018) and Makransky, Terkildsen, and Mayer (2019) and measured throughout by self-report.

⁶The discomfort belonged overwhelmingly to the Google Cardboard rather than the Meta Quest 2: asked which version was *least* educationally beneficial, 40% of students named the Cardboard against only 5% for the Quest (“Phenomenological Evaluation” 388). Among the students who judged the 2D videos most beneficial, 38.9% cited physical discomfort—disorientation, the weight of the headset over long viewing periods, the difficulty of wearing glasses inside an HMD—a figure that pools the two VR formats (389). Our own judgment in the article is that some of these issues “would have been less relevant” had students used the Meta Quest 2 rather than the Cardboard, “which is a relatively horrible experience by comparison” (390).

⁷Among the students who judged the 2D videos most beneficial, 44.5% named ease of use as the ground: convenience and accessibility of access, and the quality-of-life controls the VR formats lacked—variable playback speed, better volume and playback control, and the ability to zoom in on the footage (“Phenomenological Evaluation” 389). The rest of the students’ reasoning is broken down as follows: 11.1% cited the inability to take notes while wearing an HMD, and 5.6% had never tried a VR version at all (389).

⁸Other than the occasional in-class use of the VR headsets, students were not required to use any particular format and were allowed to use whichever of the four versions of the video content they wanted: YouTube, the PC version, Google Cardboard, or Meta Quest 2.

⁹Our inability to render a verdict places the pilot within a larger evidential pattern. Comparative studies that hold content constant repeatedly find that added immersion raises felt presence without raising learning: Makransky, Terkildsen, and Mayer, porting an identical science simulation from desktop to headset, measured more presence and less tested learning; Parong and Mayer’s content-matched comparison found a slideshow outperforming immersive VR on factual learning even as VR won every affective measure; and Mayer, Makransky, and Parong’s review of two decades of media comparisons finds no reliable learning advantage for immersive VR as a medium.

as a place they are in, and co-disclosedness, the same world opened to students as a place they are in together. The third is veri(dis)similitude: the requirement that a virtual world be similar enough to what it renders to be entered as real, yet different enough from its source material to justify its existence. A virtual learning environment, in these terms, is not a delivery medium but a manufactured and curated rhetorical ecology—an environment authored, in every element, toward a single pedagogical end, whose design succeeds when both registers of disclosure open and rhetorical incorporation is achieved.

After the pilot, we moved the course to a 2D-PC primary deployment; while students cannot experience stereoscopic 3D video without a VR headset, the videos in this interactive space are still 180- and 360-degree presentations. Our primary reason for moving away from the 3D stereoscopic VR development was due to logistics, feasibility, and necessary scale. This class needed to be made available to undergraduate students across the entire University of California (UC) system, with class sizes up to 120 students each offering. Acquiring and distributing 120 Meta Quest headsets to each student every quarter was simply not possible. Secondly, we chose to prioritize the Windows PC-based version first as it was the largest student platform population (compared to Apple laptops).¹⁰

As just noted, this class—BME 179: Biomedical Engineering Design: Addressing Unmet Clinical Needs, a one-unit online course—is offered systemwide to undergraduate biomedical engineering (BME) students across the UCs.¹¹ Students arrive with different course schedules, and the class is open across two academic calendars—Berkeley and Merced run on semesters while the rest of the universities are on quarters, though no students from the two semester campuses have enrolled in the terms taught thus far—so a fully asynchronous online format is not a convenience of choice, but the condition of the class’s systemwide existence, as a synchronous version would be nearly impossible to logistically organize. An in-person version was never possible at all, and cohort-wide scheduled simultaneity was never in the design space: the deployment maximized the only availability there was.

These constraints are more than administrative history; they define what kind of thing the virtual learning environment is. The virtual world was built, in the first place, because the actual clinical world admits so few—the operating room’s structural scarcity was the problem, and universal access was the solution the virtual environment promised. A virtual world that students cannot enter would therefore fail in a specific and instructive way: it would reproduce, in software, the very exclusion it was built to overcome. Availability is thus not a logistical afterthought but the primary pedagogical property of this course and VLE: before a virtual world can show, teach, or immerse, it must be enterable, and whatever a student cannot enter can disclose nothing to them. In the vocabulary of the actualization chain, this is a condition at A_0 , the chain’s threshold: perception, the charged image, committed judgment, and the act are all stations a student can only reach inside a world they occupy, so a world that cannot be entered does not engage its students poorly—it never begins to engage them at all. Availability is the threshold’s deployment face: the form A_0 takes when the world must be downloaded, installed, and run before it exists for its student. Nor is the scaling wall we hit peculiar to this course. Immersive VR in higher education remains largely at the prototype stage—experimental and development work rather than a regular part of teaching (Radianti et al. 1, 22)—a pattern that will be discussed in greater detail later. One further consequence of the deployment should be noted now and developed later: a fully asynchronous format changes the time of the course. There is no shared class hour; each student meets the virtual world on their own schedule and at their own pace. The VLE itself is built for company—it is multiplayer, with spatial proximity voice chat—and we clearly inform students of this capability and encourage them to meet there in their assigned groups. But with no scheduled hour to gather them, convening is left to each group’s own initiative, and the default is that a student enters the virtual hospital alone. Whether students take up the invitation to meet—and what solitary viewing does to their experience of the virtual world when they do not—are questions the three terms of survey responses will answer.

Set side by side, the pilot and the field leave a precise question standing. 70% of our pilot cohort felt more present in the headset, yet 40% judged plain 2D video the most educationally beneficial format (the largest share of the cohort to settle on any single format), and the loaner headsets sat untouched in the library; the comparison literature, holding content constant across media, finds the same shape at scale—added

¹⁰While we prioritized development for the PC version, we are currently in the process of developing a version for Apple computers, and development for VR headsets will commence once the Apple version is operational.

¹¹This class is available to undergraduate BME students at every UC except the University of California, San Francisco (UCSF), as UCSF is exclusively graduate-level health sciences.

immersion reliably raises presence and just as reliably fails to raise learning, while presence itself proves attainable on an ordinary monitor. Presence, in short, is neither scarce nor sufficient. What, then, must a virtual learning environment add to presence before the world it renders earns its place in a curriculum? What would a student gain inside the virtual world that the same footage, one click away on YouTube, cannot give them—and what kind of analysis could even name that difference?

The VR-pedagogy tradition has measured presence with increasing precision while conceding it cannot say what presence is for; the social-presence tradition has known since Short, Williams, and Christie’s founding work in 1976 that the felt company of others carries satisfaction and learning, but it matured in distance education—discussion forums, video sessions, text—around media that were channels rather than places. The two research programs descend from the same 1976 origin and cite one another almost never, having built their houses in different fields around different objects. There is a material reason the divide has persisted: until recently, few deployed pedagogical virtual environments had a social interior. Multiplayer capability and spatial proximity voice chat—the features that make a virtual world a place where students can be together—have been rarely implemented. Where educators have taught inside socially capable virtual worlds in relatively recent times—the Second Life classrooms documented by anthropologists and education researchers offer one example—the world has often been a general-purpose social platform adapted to teaching rather than an environment authored for a curriculum.¹² A desktop VLE of the present kind therefore sits at a junction neither literature has occupied: an explorable virtual hospital authored for one clinical curriculum, deployed for credit across a university system, its students group-assigned and equipped with in-world proximity voice chat—at once a rendered world of the kind the VR-pedagogy literature measures and a social space of the kind the social-presence literature theorizes. The VR-pedagogy studies compare media and score presence but do not ask what being-together inside the world does; the social-presence studies measure felt community in forums and video sessions but not inside a navigable world; to our knowledge, no analysis has read a single deployment with both sets of questions in hand. Reading one requires a frame built for the junction, and that is what this dissertation’s rhetorical-ontological and phenomenological frame supplies. Where the field’s instruments report that presence rose or interest fell, they cannot say how a wholly authored environment solicits a student’s desire, carries it toward action, or falls silent before it; the actualization chain names those moments. Where presence names a feeling, world-disclosedness and co-disclosedness name the two registers in which a virtual world is opened to its inhabitants—as a place I am in, and as a place we are in together—so that what a design achieves or forfeits can be located, not merely scored. A VLE, on this reading, is a manufactured and curated rhetorical ecology with a single pedagogical end, and its effectiveness is a design question in the strictest sense: the environment teaches when it is designed so that both registers open—when rhetorical incorporation is achieved.

The survey given to all students across the Winter 2025, Spring 2025, and Winter 2026 quarters was designed, fielded, and collected by myself and Prof. Christine King—the instructor of record—and is read here through the rhetorical-ontological and phenomenological frame this dissertation has been building. The data yield the findings, with every count and quoted response coming from the survey corpus. The theoretical frame supplies the how and the why: the actualization chain and the methodology of Rhetorical-Ontological Diachronic Analysis (RODA) serving as a background grammar for interpreting students’ responses—which moment of engagement is in play when a student reports presence, confusion, the flight to YouTube, or the desire for more socially interactive course design. Accordingly, what follows moves through seven parts. The first reviews the relevant literature in its two traditions: the desktop route as a pedagogical modality in its own right, with presence relocated from hardware to the design of the environment; the comparative evidence that more presence does not deliver more learning; the rationale for asynchronous deployment at scale; and the research on learning together in shared virtual space—social presence and computer-supported collaborative learning—whose near-total lack of contact with the VR-pedagogy literature is itself a finding this section works across. The second part describes the VLE and its deployment: the virtual hospital as a designed world, the cohort of 241 students across the three quarters, and the survey instrument together with the standing limits of what it can show. The third presents the survey findings in order: the threshold

¹²On the education-research side, the fullest example in this section’s literature review is Merchant et al.’s structural-equation study of college chemistry instruction inside Second Life (N=204), taken up in detail below. On the anthropological side, Tom Boellstorff—an anthropologist at UC Irvine—conducted multi-year ethnographic fieldwork inside Second Life through the avatar Tom “Bukowski,” the fieldwork behind his monograph *Coming of Age in Second Life*; he recounts that work in “Virtual Worlds and Futures of Anthropology” and carries the program forward to the present platform generation in “Toward Anthropologies of the Metaverse.”

problems that kept some students out of the virtual world; what students found, and could not do, once inside; the flight to YouTube; the presence students attested on ordinary screens; the group co-watching capability that was had, wanted, and largely unactivated; and the shift in student skepticism from technical friction to the question of the platform’s value. The fourth part conducts the rhetorical-ontological and phenomenological analysis: it locates the primary rupture point in the student experience, reads the boredom the data attest through the dissertation’s three-form account, takes up the justification question the data raise, and shows how the VLE is world-disclosed and co-disclosed to its students. The fifth turns diagnosis into design, ranking the design levers by the moment of engagement each one works on, with the activation of the WE given the fullest treatment. The sixth states the limits of what this survey data can unconceal. The seventh names what the analysis hands forward—to the boredom analysis, a separate section that follows this one, and to the field.

By reading the three quarters of survey data through this frame, the following analysis will substantiate a claim with three connected parts. First: the deployed desktop world succeeds where our pilot said screens could succeed. Across three terms, students attest to presence on ordinary monitors at rates matching what the pilot measured in VR headsets; a virtual world on a flat screen can still place its student somewhere, and world-disclosedness does not require a headset. But presence, I will show, is not the finish line. As deployed, the virtual hospital gives students a great deal to see—filmed operations, physician and engineer interviews, explorable procedure rooms—and watching is nearly all it gives them to do. Watching is itself an act: a student can stand in the virtual operating room and refuse to attend, and the one who watches has actualized a potentiality the world furnished. The shortfall lies in what else the world makes available: no task to carry out inside it, no procedure to attempt, no problem whose solution it asks of the student—no richer or more pedagogically effective potentiality for the student’s engagement to actualize. In the chain’s terms, the deployment carries engagement to committed judgment (A_3)—the world wins the student’s attention and their assent that it is worth taking seriously—but the only completed act (A_4) it affords that commitment is one the student could have performed anywhere: watching. The world falls silent at the passage from A_3 to A_4 not because no act follows, but because the acts it affords stop at watching. Second: because every video housed in the virtual hospital also exists one click away on YouTube, the desktop world carries a standing burden of justification. It must be different enough from the video of itself to justify existing, and that difference can only be made of what video cannot render: a place to be in, a task to do, and others to be there with. Third: of those three materials, the last is the one this deployment alone possesses. Watching together in the same virtual space, within earshot of one another, is a capability no other route in this program’s history has offered—the published VR version had no real-time collaborative features at all, and students could only enter it alone (“Phenomenological Evaluation” 389). Where place and task deepen world-disclosedness, being there together opens a different register altogether—co-disclosedness: the virtual world disclosed not to each student alone but to a WE. I will argue that this capability is at once the desktop world’s deepest answer to “why not just YouTube?” and, as the data demonstrates, its deepest unrealized asset. The scope of this claim is a matter of development history rather than principle: nothing in a headset forbids company; it is on the desktop route that company was built. What the analysis owes, then, is evidence for each part: that the world was disclosed, that the acts it furnished stopped at watching, and that co-disclosedness—the one capability that could justify the world—went largely unused.

2 The Two Literatures

The desktop route has a research literature of its own, and that literature’s founding model already refuses the reading a headset-first field invites: the desktop build as a lesser headset. Dalgarno and Lee’s model of learning in three-dimensional virtual environments locates what is pedagogically distinctive about the medium in what an environment lets its learner do and see—learner interaction first, representational fidelity alongside it—and treats presence and co-presence not as endowments of hardware but as consequences of those characteristics: in the model’s words, “a consequence of the representational fidelity and learner interactions facilitated by the environment” (“Learning Affordances” 14). The affordances the model derives are, accordingly, task-possibilities in Gibson’s sense—things an environment makes available to an agent to do—not display achievements. The model arrived carrying its authors’ own verdict on the field’s evidence: “[m]uch of what has been published about the educational uses of 3-D technologies is largely ‘show-and-

tell’, presenting only anecdotal evidence or personal impressions that cannot be usefully generalised beyond the local context” (23). When the same authors put the model to its own test two years later—surveying fifty-three higher-education staff on which afforded learning tasks had yielded which of the five predicted benefits—respondents endorsed every benefit at nearly uniform strength regardless of task, none of the hypothesized task-to-benefit links reached significance, and the authors flagged the deflating explanation themselves: the uniform agreement “could be more a reflection of their enthusiasm and positive attitude towards the use of 3D VLEs than an evidence-based assessment of the actual learning benefits” (“Afforded Tasks” 243). The field’s central model thus stands named beside its own inconclusive self-test.

The model’s sharpest critique asked not for more data but for pedagogy: a decade of educational-VR research had run technology-first, deriving expected benefits from technical affordances with almost no explicit account of how learning occurs.¹³ What Fowler’s repair adds, and what this section will hold onto, is a curve: “the assumption that high levels of representational fidelity and learner interaction will result in deeper learning is questionable. It may be argued that there are optimum levels of these characteristics that maximise learning. Going beyond the optimum increases costs and results in a limited or even negative return with respect to learning benefits” (Fowler 415). Neither characteristic improves learning in proportion to its own increase: each has an optimum level, and beyond it, added fidelity or added interaction yields limited or even negative returns for learning. That caution matters most for representational fidelity, because fidelity is the characteristic that headset hardware exists to advance—and the one a headset-first field kept mistaking for the medium’s pedagogical substance. Our own pilot had already produced a concrete instance of that curve. We expected that students’ familiarity with VR and video games would incline them toward the virtual-reality versions; the correlation ran the other way: of the students who judged the plain 2D videos most educationally beneficial, 66.7% were weekly gamers, against 33.3% of those who favored the VR versions—and our interviews traced the judgment to builds that “do not meet the production standards of contemporary VR and traditional video games” (“Phenomenological Evaluation” 389). We stated the result then at its proper strength—a possible correlation, awaiting validation (389)—but its direction is the one the curve predicts once experience is allowed to move the optimum: the students most practiced in polished, feature-rich worlds hold the highest bar, and a build that undershoots it loses them first.

Within the desktop medium itself, the process studies converge on where the leverage lies, and it is not the display. Makransky and Petersen, modeling 199 medical undergraduates through a desktop genetics simulation, found two routes from the medium’s features to knowledge gains—“the affective and the cognitive paths” (“Investigating the Process” 24): presence feeding motivation on one, perceived usability feeding cognitive benefit on the other, the two converging on self-efficacy as the sole direct predictor of learning.¹⁴ In Merchant and colleagues’ semester of Second Life chemistry (N = 204), the gate is usability: “the 3D virtual reality features can support the development of learners’ spatial orientation ability, self-efficacy, and presence only when the learners’ [sic] perceive the experience as meaningful and the system easy to use” (Merchant et al. 562).¹⁵ Dubovi, Levy, and Dagan’s nursing simulation, finally, returned the literature’s flattest correction to the hardware account of presence: on ordinary monitors, their students’ presence scores met or exceeded the norms the construct’s founding instrument had recorded for immersive VR,¹⁶ and they drew a two-part conclusion whose parts deserve different fates: “presence is not a unique attribute of the environment, but it is induced by the fidelity of representation and the high degree of interaction and user control” (Dubovi et al. 25).

The first clause is right, and the survey data I will analyze later on confirm it at scale: presence is not what headsets have and screens lack. The second clause gives fidelity a role the evidence does not sustain—not the wider field’s evidence, and not this study’s own. The one component of their presence measure that correlated with learning was involvement and control ($r = 0.211$ overall, 0.231 on the procedure’s core

¹³The ten-year review behind the charge found explicit learning theory almost wholly absent from the literature it surveyed: “[a]ll the other reviewed articles do not refer explicitly to a learning theory” (Mikropoulos and Natsis 775, qtd. in Fowler 412). Fowler’s repair works outward from intended learning outcomes to the technology that would serve them, rather than inward from the technology’s affordances (412–13, 416–20).

¹⁴Self-efficacy $\rightarrow$ learning, $\beta = 0.579$; presence’s effect reaches learning only through motivation and then self-efficacy, never directly (“Investigating the Process” 23–24).

¹⁵The mediation is nearly total ($\beta = 0.956$, VR features $\rightarrow$ usability; Merchant et al. 560). The design is single-condition desktop—no headset arm—so the study licenses no claim about media sufficiency in either direction.

¹⁶Involved/Control 5.2 ± 0.7 , Natural 4.7 ± 1.03 , Interface Quality 5.0 ± 1.1 , against Witmer and Singer’s immersive-VR norms of 5.2 ± 0.8 , 4.1 ± 1.1 , and 4.8 ± 1.1 respectively (Dubovi et al. 23).

subscale); the realism-adjacent components tracked nothing (25).¹⁷ Presence flourishes in worlds whose fidelity is schematic or frankly poor—the cube-built landscapes of *Minecraft*, the aging textures of *World of Warcraft*—worlds no fidelity-led account explains. What induces presence is not the exactness of the copy but the constitution of the artifact as a world: an environment built as an ecology, one that answers action—a world the student moves and is moved by in turn. Fidelity is one tool for building such a world. It is not the ground of it.

Presence itself entered the research literature as a property of persons, not of headsets. Lombard and Ditton’s founding synthesis defines it as “the perceptual illusion of nonmediation”—an illusion that “occurs when a person fails to perceive or acknowledge the existence of a medium in his/her communication environment and responds as he/she would if the medium were not there”—and the definition is deliberately medium-general, built to hold across television, film, and virtual reality alike (“Presence Explicated”). The consequence is drawn in the same section: “[b]ecause it is a perceptual illusion, presence is a property of a person,” varying with a medium’s form, its content, and the person attending to it (“Presence Explicated”). The founding instrument agrees about the location. Witmer and Singer, constructing the field’s first validated presence questionnaire, reject the view that immersion is an objective property of the display hardware: “immersion, like involvement and presence, is something the individual experiences” (227). Their paper is equally plain about what had not been shown: factors that increase presence were expected to increase learning, “but as yet we have no direct evidence to support our contention” (239). Two decades of determinants research kept resolving the question user-side—in Chow’s structural model of a desktop virtual world, computer self-efficacy, the student’s confidence with the machine, was the strongest predictor of presence, ahead of any property of the display (11–13).¹⁸ Across research traditions that rarely cite one another, the convergence is the same: presence is achieved by persons and occasioned by worlds; no hardware tier owns it.

The media-comparison studies of the past two decades tested the direct question: does the added presence of a more immersive medium produce more learning? Across content-matched designs, the answer was no. Makransky, Terkildsen, and Mayer ported an identical laboratory simulation from desktop monitor to head-mounted display: presence rose sharply in the headset ($d = 1.30$ in the first lesson, 2.20 in the second), tested learning fell, and EEG measures located the mechanism in cognitive overload (“Adding Immersive” 232–33).¹⁹ Parong and Mayer ran the media comparison with content matched line for line: a narrated slideshow beat immersive VR on the knowledge posttest in both experiments while VR won every affective measure taken—enjoyment, motivation, engagement, interest (“Learning Science”).²⁰ Their mediation follow-up closed the causal loop: the medium’s effect on retention ran through self-reported extraneous load and high-arousal positive affect—the features that make immersive VR appealing are the ones implicated in its learning cost (“Cognitive and Affective Processes” 236–37). Mayer, Makransky, and Parong’s review of the program’s two decades states the upshot plainly: “[t]here is not strong evidence that students learn academic material more deeply when it is presented in IVR than when it is presented with conventional media” (2236). Barrett and colleagues sharpen the point from the psychophysics side: on one category-learning task, learning accuracy was statistically equivalent across VR, desktop-3D, and 2D versions; what differed was response time, fixation behavior, and—in VR alone—attrition from physical discomfort (12–14).²¹

The health-professions simulation meta-analyses locate where screen-based practice does reliably improve learning: in skills and clinical reasoning rather than declarative knowledge. Kononowicz and colleagues’ meta-analysis of screen-based virtual patients found nothing added on knowledge over traditional formats ($SMD = 0.11$, non-significant) and a moderate-to-large advantage on skills and clinical reasoning ($SMD = 0.90$) (6–7). The base rate behind that split is settled: across 609 studies of technology-enhanced simulation, pooled effects against no intervention are large on every outcome family, and the reviewers close

¹⁷Their own design principle, stated two sentences before the fidelity clause, names control alone: “ensuring that participants gain sufficient control of the environment is an important condition for learning” (Dubovi et al. 25). The fidelity clause is carried by citation to prior literature, not by the study’s own data.

¹⁸Chow’s model is scoped to self-presence rather than spatial presence, and his own design implication follows the data: invest in orientation and user confidence rather than in richer rendering alone (12–13).

¹⁹The dissociation also holds inside a single desktop condition: in Merchant and colleagues’ chemistry model, usability generated presence ($\beta = 0.540$) while presence’s path to learning was null ($\beta = 0.069$, $p = .367$; Merchant et al. 560).

²⁰The slideshow’s advantage on the knowledge test given after the lesson was $d = 0.92$ in the first experiment and 1.12 in the second. The same study carries the constructive result: a generative summarizing prompt inside VR closed the learning gap without reducing enjoyment—the deficit is remediable by instructional design, not intrinsic to the medium.

²¹The equivalence claim is accuracy-specific, and the 2D arm is small ($n = 8$)—a scope limit worth carrying with the citation.

the question themselves—“we question the need for further studies comparing simulation with no intervention” (Cook et al. 1987).²² Simulation teaches; the evidence thins exactly where a world is only watched. The knowledge/ skills split names where the medium’s value lives: not in what a student sees from inside it, but in what the student is given to do there.

Read together, these two decades leave a precise explanatory hole. Presence belongs to the person; presence does not convert into learning on its own; learning gains concentrate where the environment gives its student something to do. The field has measured all three results and has no construct that joins them—no name for the difference between being placed in a world and being taken into it as an agent whose acts complete what the world offers. Game studies built that construct: in Gordon Calleja’s *In-Game: From Immersion to Incorporation*, incorporation names the double movement in which the virtual environment is absorbed into the player’s consciousness while the player is embodied within it as an agent. The education literature does not import it; the bridge between the two fields remains unbuilt in both directions. Part II rebuilt the construct on this dissertation’s own ground—rhetorical incorporation, one in substance and dual in being, world-disclosedness and co-disclosedness—and the desktop deployment now gives it empirical work to do. The survey data show a world that places its students: presence attested on ordinary monitors, exactly where the field’s own findings say it can live. What the data must then answer is whether the world completes what it offers—and that is a question about rhetorical incorporation, not about presence, which is why the frame Parts I–II built is the instrument this section reads with.

Whether an at-home, unsupervised deployment gives up anything a supervised classroom would provide has been tested directly. Makransky, Mayer, Veitch, and colleagues had 112 students complete the same desktop science simulation either at home or in class: learning outcomes did not differ ($d = 0.14$), nor did intrinsic motivation ($d = 0.02$) or self-efficacy ($d = 0.09$), and the null pattern survived adjustment for prior knowledge and goal orientation (“Equivalence” 8–9).²³ Their interpretation extends Clark’s old verdict on media to physical context: the instructional method—the simulation itself—drives the learning, and felt presence is proposed, though not tested, as what renders the location unimportant (10). The scope of the design deserves equal attention. Both groups worked with the identical desktop simulation—a Labster-built virtual biology laboratory—and only the setting changed: one group at home, unsupervised, the other in a classroom with a teacher present. No group received a different medium. The study therefore answers one question and leaves a second untouched. It shows that moving a desktop simulation out of the classroom and into the student’s own home, on the student’s own schedule, costs nothing measurable in learning, motivation, or self-efficacy—the fact an asynchronous online course needs established. It does not show that an interactive 3D simulation teaches better than a flat video of the same content, because no video condition existed in the design. That second question is the one our own deployment poses: every term, part of the class entered the interactive desktop world while another part watched the same operating-room footage on YouTube instead, and the survey data speak directly to what separated those two experiences.

The field’s own deployment history tells the complementary story. Radianti and colleagues, reviewing 38 immersive-VR applications in higher education, found that “the majority of articles indicated that VR for education is still in its experimental stages—prototyping and testing with students,” with evaluation skewed toward usability rather than learning and the maturity level itself “a barrier for its adoption in regular teaching activities” (22).²⁴ A virtual world in routine use every term, across a university system, is what that literature documents almost never happening; the desktop build is what makes the routine part attainable. The headset route also carries a physical cost the desktop route does not levy. Cossio and colleagues’ review of 25 HMD studies reports a mixed rather than uniform picture—twelve studies with inconsistent or very low cybersickness scores, five with high scores, oculomotor disturbance the most common symptom—but where head-mounted delivery was compared directly against desktop or non-immersive formats, the worse outcomes fell on the headset side, “likely due to its more immersive nature and the potential for greater

²²The two meta-analyses do not overlap: Cook and colleagues pool technology-enhanced simulation broadly and explicitly exclude the screen-based virtual patients Kononowicz and colleagues analyze. The virtual-patient meta-analysis of that earlier era is Cook, Erwin, and Triola (2010), a different study, not drawn on here.

²³One interaction survived adjustment: mastery orientation predicted self-efficacy more strongly in the classroom group (“Equivalence” 9–10). The sample is self-selected (112 of 289 eligible students consented), a limit the authors flag themselves (11–12).

²⁴The review’s inclusion criterion is head-mounted immersive VR only—desktop VR is excluded by design—so the prototype-stage verdict describes precisely the hardware route, not the desktop one (24).

sensory conflict” (14).²⁵ Three findings from this literature bear directly on the shape of our course. Students learn as much from a desktop simulation completed at home as from the same simulation completed in a supervised classroom (“Equivalence” 8–9). Headset VR in higher education remains, by the field’s own account, a prototype technology—built and tested with students, only rarely made part of a regular course (Radianti et al. 22). Headsets carry a documented cybersickness cost that ordinary monitors do not (Cossio et al. 14). Our course is fully online and asynchronous, serving undergraduate biomedical engineering students across the University of California’s campuses, each on their own machine and their own schedule. For a course of that shape, the desktop deployment is not a diminished substitute for headset delivery. The home-versus-classroom evidence says our students lose nothing by working where and when they want. The deployment reviews say the headset alternative has almost never been made routine at any scale. The cybersickness literature says the monitor spares them a physical cost besides. The desktop build is what the field’s own evidence recommends for the course we actually teach.

The section’s second literature studies a different presence: not the sense of being in a place but the sense of being with someone. Social presence entered the research vocabulary in 1976, in Short, Williams, and Christie’s social psychology of telecommunications, naming how fully a communication medium lets its users register one another as persons—decades before rendered worlds existed to be present in. Biocca, Burgoon, and Harms, surveying the construct’s first quarter-century, define its core as “the sense of being with another” (456) and set out the criteria a robust theory of it must meet. Their sharpest verdict falls on the field’s own instrument. The inherited scale was built to rate media channels, not to describe interactions, and the authors press the question that follows: “is social presence just an attitude towards a medium?” (469).²⁶ A construct meant to describe an interaction had been operationalized as an opinion about equipment.

What the construct predicts, in its home tradition of online education, is substantial. In Gunawardena and Zittle’s origin study of a text-only academic conference, social presence alone accounted for roughly 60 percentage points of the variance in learner satisfaction (18–19).²⁷ Richardson and Swan, adapting the same scale, found social presence correlating .68 with perceived learning (73–74)—a second result from the same research program, extending rather than independently confirming the first.²⁸ The meta-analytic aggregate holds the pattern at scale: across 26 effect sizes, social presence correlates 0.56 with satisfaction and 0.51 with perceived learning (Richardson et al. 410–11).²⁹ Terry and Doolittle press hardest on what the construct has become: the Community-of-Inquiry tradition entangled social presence with the teaching and cognitive presences around it, and interactivity keeps being mistaken for it. The repair they propose is to identify it “as a psychological state, as opposed to one that is mediated through, or as a by-product of the use of a specific technology” (121).³⁰ That relocation is this section’s load-bearing result: social presence, like spatial presence in the first literature, belongs to persons. It also sets the construct on ground Heidegger prepared: being-with-others is not an addition to a self-standing subject but a constitutive structure of human existence—the world is always already one shared with others—and the distinctive enactment of that structure is speaking-with-one-another.³¹ A design that opens room for speaking-with-others therefore does more than decorate a channel: it furnishes the enactment in which others are there at all. It is something a

²⁵The aggregate is a reportable moderator, not a uniform verdict: Huber and colleagues (2018) found cybersickness *less* prevalent in fully immersive VR (3.3%) than in augmented reality (6.6%) (Cossio et al. 14). The same cost surfaces as the VR-only discomfort attrition in Barrett and colleagues’ three-arm comparison.

²⁶The scale’s provenance explains its orientation: it descends from studies funded by the UK Post Office, the Department of Transportation, and General Electric—organizations commissioning ratings of their communication channels (Biocca et al. 469).

²⁷Their closing design recommendation cites Johansen, Vallee, and Spangler’s suggestion that social presence can “be cultured”—built up by design and practice rather than fixed by the medium (Gunawardena and Zittle 23).

²⁸The correlation is strongest in class discussions (.83) and group projects (.80), weakest in lectures, notes, and readings (.40) (Richardson and Swan 74–75).

²⁹Both correlations are moderated by course length, discipline, and by which scale was used—no two social-presence scales are equivalent (Richardson et al. 410–12, 415). The outcome throughout the meta-analysis is self-reported perceived learning, never graded performance (411); Tu’s decomposition of the construct returned reliabilities he himself calls “medium-low” and an instrument offered as “a first step” (Tu 43–44).

³⁰Five factors contribute to it on their account: group cohesion, co-presence and the salience of others, affective association, immediacy, and intimacy (Terry and Doolittle 130–31).

³¹“The world of Dasein is a *with-world* [*Mitwelt*]. Being-in is *Being-with* Others. Their Being-in-themselves within-the-world is *Dasein-with* [*Mitdasein*]” (BT 155; italics Heidegger’s). The Marburg lectures on Aristotle name the enactment: “The distinctive mode of being in being-with-one-another is in speaking-with-one-another. To set forth the possibilities of being-with-one-another is the *ἔργον* of rhetoric” (*Basic Concepts* 91–92).

design can activate, not something a medium contains.

The same tradition states what activating it requires. Garrison, Anderson, and Archer locate social presence in the communication context rather than the medium and name teaching presence “[t]he binding element in creating a community of inquiry” (96)—when text-based education fails, their diagnosis is a deficit of teaching presence, not of bandwidth. Kreijns, Kirschner, and Jochems give the tradition its bluntest warning, naming the first of two pitfalls of computer-supported collaborative learning: “taking for granted that social interaction automatically takes place just because an environment makes it technologically possible” (336)—the second being the restriction of what interaction does occur to task-focused exchanges. Providing the channel is not producing the contact; sociability must be designed for. The findings extend into rendered worlds. De Back, Tinga, and Louwerse’s immersed collaborative environments produced learning gains with an inverse group-size gradient—smaller groups, larger gains (5375–76).³² Van der Meer and colleagues’ review of 139 virtual-reality collaborative-learning systems found 78.4% of them monitor-based against 7.2% head-mounted (6).³³ Collaborative learning in virtual worlds, as actually practiced and studied, already runs on ordinary screens.

These two literatures share their origin and almost nothing after it. Short, Williams, and Christie stand at the head of both: the presence tradition the VR-pedagogy field inherited and the social-presence tradition of online education descend from the same 1976 book. Downstream, the citation traffic stops.³⁴ This section reads a desktop world whose students meet inside it by voice, and so it must join two bodies of evidence that have not met each other—the second literature’s vocabulary applied to an object the first literature built. What this movement hands forward is the base for that work: a construct that names being-with as its own register of presence; evidence that it predicts satisfaction and perceived learning; the warning that it must be designed for and convened, never assumed; and the finding that it belongs to persons—available, in principle, to any world that gives its students a way to be there together.

3 The World and Its Deployment

The deployed platform is a downloadable application for Windows PCs, built in Unreal Engine, that renders a navigable virtual hospital.³⁵ Students move through it in the first person, by mouse and keyboard, on their own monitors. Each procedure room gathers the materials for one operation or category of operations. The room’s principal material is the operating-room footage, which was captured two ways at once: a 360-degree stereoscopic camera records the whole room, and a head-mounted camera worn by the operating physician records the surgeon’s own first-person view (“Scalable Virtual Reality” 4). Each room also holds recorded interviews with the physicians and engineers involved and supplementary lectures and readings (4), together with animated models of the procedure or its equipment. The filmed procedures now span multiple institutions and countries. Seven procedures were recorded at the UCI Medical Center and the Children’s Hospital of Orange County: bilateral scar revision, transcatheter aortic valve replacement, deep brain stimulation, spinal deformation surgery, pediatric auricular reconstruction, cornea transplant, and standard and laser cataract procedures. A perimortem C-section simulation was recorded at the UCI Medical Education Simulation Center. To these the platform adds standard cataract procedures filmed in hospitals in Vietnam and Paraguay, placing the same operation side by side in three healthcare systems (3–4). On an ordinary monitor the footage plays as an interactive 360-degree panorama; the same materials are mirrored to a YouTube channel and website for students whose machines cannot run the application (4).

The world is built for company. Voice chat is real-time and spatial—students hear the classmates near them in the virtual space—and a group can convene inside the environment, watch a procedure together,

³²Cited for collaborative-learning design and outcomes only; social presence does not appear in the study’s text.

³³Among the nine design features the review requires of such environments, verbal audio communication is judged essential and support for socialization is named a primary function (van der Meer et al. 9–11).

³⁴The meta-analysis that pools the social-presence evidence draws its studies from online courses, not rendered worlds, and pools neither Biocca nor Kreijns (Richardson et al. 415–16); the desktop and headset studies of the first literature measure spatial presence, cognitive load, and learning outcomes, and the social-presence instruments appear in none of them.

³⁵The platform facts in this part are reported in a second of my publications, co-authored with Prof. King and Kadin Diec, and cited parenthetically by abbreviated title hereafter: King, Christine E., Kadin Diec, and Dalton Salvo. “Scalable Virtual Reality Global Clinical Immersion for Culturally Responsive Engineering Design Skills.” 2025 ASEE Annual Conference & Exposition, paper #49715. The paper has no journal pagination; parenthetical page numbers refer to the PDF.

and share various other artifacts through the screen sharing functionality: the design “supported real-time voice chat and screen sharing, enabling collaborative team viewing if students wanted to meet in groups within the virtual environment” (“Scalable Virtual Reality” 4). That design choice answers the firmest requirement the collaborative-learning literature states: in van der Meer and colleagues’ review of 139 virtual-reality collaborative-learning systems, “efficient communicational tools are essential for effective collaboration: verbal (audio) communication is crucial” (9). The voice channel, that is, is not one feature among others; it is the medium of speaking-with-others, the mode in which a group is a group at all.³⁶ Every student is assigned to a procedure-paired group at the start of the term, the capability is announced in the course materials, and meeting in the world is encouraged. Group interaction inside the platform is one of the course’s stated practice objectives (3). Co-watching in the virtual hospital is thus not an incidental feature but the designed use of the environment’s social architecture.

Every element of this environment is authored. The hospital and its rooms, the two camera vantages, the interviews, the group assignments, and the channel over which students speak are design decisions, and each is a made source—an *archē*—of the experience it occasions. The virtual hospital is, in the dissertation’s terms, a manufactured and curated rhetorical ecology: a designed *Umwelt* whose makers supply the matter and the form of the encounter while the act remains the student’s own. The inventory above is therefore also an inventory of design levers, and the analysis will return to each element as the survey data show what it did and did not occasion.

The survey corpus comes from three offerings of the course: Winter 2025 (102 respondents), Spring 2025 (79), and Winter 2026 (60)—241 students in all. Each term used the same 14-item end-of-course instrument, four items of which were open-ended; those four items produced the 873 open-text responses analyzed in this section. One wording note applies throughout: the instrument asks about the “VR platform,” phrasing inherited from the pilot era, though the deployment it surveys is the 2D-PC-primary build described above. The instrument also recorded whether students could run the application and whether the platform elicited immersion, and those answers divide each cohort into two strata: students who used the desktop application (52%, 51%, and 50% across the three terms) and students who watched the videos only (45%, 46%, and 43%). The split is statistically indistinguishable across terms ($\chi^2 = 0.02$, Cramér’s $V = 0.009$, $n = 231$), and that stability carries weight in what follows: it licenses reading differences between the strata as structural features of the deployment rather than accidents of a given cohort. The course survey is itself anchored in the global deployment: its title is “Global Healthcare Needs and Final Assessment Survey,” and its three content items test the global material—nearly all students answered them correctly. The dedicated global-procedures questionnaire—two Likert items on exploring healthcare and on seeing cross-country differences, and an open invitation to comment on the Vietnam and Paraguay videos—was a separate Winter 2025 administration, reported in the ASEE paper above; two of its items share their wording with this survey’s platform-use and immersion questions. Where its results are used, they are cited to that paper.

Responses were anonymized before analysis: names and identifiers were dropped, and each response carries a term-and-respondent code. From the 873 open-text responses I derived a coded corpus. The codebook was fixed before coding began, and its categories come from the dissertation’s own analytic apparatus. It contains roughly forty codes organized into eight families, each family collecting the codes for one kind of signature: forms of boredom; flights away from the platform; the texture of reported time; the station of the actualization chain a response addresses; the involvement channels a response names; signatures of rhetorical incorporation; student suggestions; and an open family for codes that emerged during coding. Coders read each response segment by segment, a segment being the clause or sentence carrying one codable idea, and recorded a set of codes per response, with verbatim evidence spans retained for every application in the load-bearing families. Coding was performed by AI coders working blind under fixed written instructions; I adjudicated every disputed application myself.³⁷ All 873 responses were coded at least once, and a 143-

³⁶Heidegger, reading Aristotle’s definition of the human being as the living thing that speaks: being-with-one-another “is not being-with-one-another in the sense of being-situated-alongside-one-another, but rather in the sense of *being-as-speaking-with-one-another* through communicating, refuting, confronting” (*Basic Concepts* 33; italics Heidegger’s).

³⁷All coding passes in the final corpus were performed by Anthropic’s Claude Fable 5 model, each pass blind and independent under written coder instructions reproducing the codebook. Three batches were initially single-coded by a smaller model (Claude Haiku 4.5) for economy; a same-instrument comparison measured its agreement with Fable 5 at mean Jaccard .51–.67 against the .878 same-model benchmark, and I discarded those passes and had the batches recoded, so no Haiku coding remains in the corpus. Per-code agreement on the double-coded subsample ranged from 50% to 100%, weakest on union-merged mention codes and strongest on the load-bearing families. Every cross-pass dispute on a load-bearing code reached me for

response subsample—drawn deliberately from the term the codebook had least seen—was coded twice by independent blind passes, yielding a mean Jaccard agreement of .878 over full code sets, the set-overlap measure appropriate to multi-label coding (Krippendorff’s α with MASI distance, .82). Merging the two passes was deterministic and asymmetric: mention-level families merged by union, the analytically load-bearing families by intersection, with the four residual disputes resolved by hand; a subsequent line-by-line review of the whole apparatus added three further manual corrections. Throughout, codes name signatures in student talk, never certified inner states.

The corpus has four standing limits. It is a single self-reported channel: no physiological, behavioral, or performance measure accompanies it, so claims about inner states rest on what students chose to write. It is retrospective: students answered at the end of the term, so the corpus preserves remembered experience, not experience in progress. It supports no learning-outcome claims: the course’s content items sat at ceiling, and nothing here measures whether one format taught better than another. Term-to-term comparisons carry a composition caveat: the three cohorts differ in size and possibly in makeup, so cross-term movements are read as trends, not tests. What the corpus does provide, at a scale the pilot could not, is 241 students’ own accounts of entering—or failing to enter—the same virtual world.

4 The Deployment Findings

The survey divides the 241 students into two groups, and the division itself is the first finding. Each student reported whether they used the desktop application or watched only the YouTube copies of the same footage: 123 used the application, while 108—nearly half of the total participants—chose the answer “I utilized the YouTube videos only.” The YouTube group is itself two stories. Sixty-one of its students reported that they could not use the application at all; for them, YouTube was not a preference but the only door left open. The other 47 reported that they could have used the application and turned to YouTube anyway—a flight resulting from hassle rather than a failure to access the platform. The complaints make the difference between the groups concrete. Among the 108 students on the YouTube route, about one in six—17%—described problems installing the application or getting it to run, and for most of them the problem was final: fifteen of those eighteen students are among the sixty-one who never got in. Among the 123 application users, only 3% mention installation at all, and theirs are accounts of difficulty overcome, since these students did, in the end, get the program running. The most common obstacle was the operating system itself: the build ran only on Windows, which excluded all students who used Apple products, leading them to want it to be “[m]ore inclusive for macbook users :”(W25-R042/Q9).³⁸ The account gate stopped others before the hospital ever appeared; one student’s entire answer to the improvement question was “[m]ake sure the sign-in is actually possible” (W26-R059/Q9). A smaller number were stopped by the machine itself, reporting hardware too weak for the application or no Windows computer to put it on (7% of the YouTube group, against 3% of application users). One of them described the strain in physical terms: “I think the VR program needs more tweaking, especially to work on a less powerful computer, because, at least for me, my laptop sounded like it was about to blow up after 5 minutes” (S25-R033/Q6).

The application users tell a different friction story, and theirs happens inside the world. Their complaints are those the analysis was designed to test. Given the development and testing of the Windows build, we predicted five elements where students might encounter friction in using the virtual space: buffering delay on the video streams, since every video is delivered over the internet from cloud storage rather than shipped inside the application; wayfinding, the work of locating a particular room in a multi-story hospital; video fidelity, the sharpness of the footage once a room’s screen is playing it; the close proximity to the screen that comfortable viewing requires; and avatar occlusion, one student’s avatar blocking another’s view in a shared room. Predicting where a designed thing will break is part of the work of design itself. The design theorist Thomas Wendt, following Gui Bonsiepe, treats breakdowns as “a necessary accompaniment

manual ruling, and I subsequently reviewed the apparatus station by station—every rule, boundary, and disputed application. A dedicated two-pass recount of the shared-involvement evidence followed the same protocol (93% exact agreement). All pass files, comparison reports, and adjudication records are retained.

³⁸Survey verbatims are cited by term, respondent, and survey item: W25-R042/Q9 is Winter 2025, respondent 42, question 9. The three terms are Winter 2025 (W25), Spring 2025 (S25), and Winter 2026 (W26); respondent numbers were assigned during anonymization; question numbers are the survey’s own item numbers. Student responses are quoted verbatim throughout, including typos and punctuation.

to design,” each one revealing “the tissue of the relations needed to fulfill the tasks” (qtd. in Wendt, *Design for Dasein* 33–34); the designer’s charge is to “anticipate the forms of breakdowns” rather than to imagine them away (*Design for Dasein* 33–34). The survey returned a verdict on each prediction. Buffering and latency drew complaints from 11% of application users (25 complaints in all); wayfinding from another 11% (21 complaints), often paired with requests for maps or clearer guidance; video fidelity from 5%, whose complaints of blurry or hard-to-see footage name the worry in the students’ own vocabulary. The predicted proximity problem barely registers—two complaints—and avatar occlusion never appears, for the simple reason that most students entered the hospital alone: with no other avatars in the room, there was no one to block the view. Beyond these five we had also anticipated frictions of a blunter kind: crashes at startup, given how demanding Unreal Engine 5 is of an ordinary student laptop; the exclusion of Mac users; and the absence of conveniences students know from YouTube—above all changing the playback speed, a feature we deliberately withheld as a pedagogical decision. All three arrived. Crashes are the largest single friction in the corpus (15% of application users; 29 complaints); the Mac exclusion did its work at the threshold; and students asked for exactly the feature we had withheld: “I think it would be helpful to have the option to change the playback speed of the videos” (W26-R021/Q9).³⁹

Almost none of this concerns the world itself. Taken together, the two friction stories name the machinery around the virtual hospital—the installer, the operating system, the account gate, the stream, the frame rate—and almost never its architecture, its rooms, or its footage as designed. For the students it stopped, that machinery decided everything: a world that cannot be reached discloses nothing, and what was lost at the threshold was world-disclosedness itself. The virtual hospital foreclosed rather than disclosed, and it foreclosed on technical grounds—not through any design aspect of the world itself. In the chain’s terms the finding touches A_0 alone: availability is in fact its condition of possibility, and for a quarter of the class the world closed at the installer, before anything in it could solicit so much as a look.

Students praised what the world gave them to see. Mentions of the content—the filmed operations, the physician interviews, the interest of watching real surgery—are the largest positive channel in the whole corpus: 166 coded mentions from 147 students, 61% of the class. The praise comes from both groups at nearly the same rate (67% of application users, 59% of the YouTube group), which follows from the deployment itself: both routes carried the same footage, and what students valued was what the footage showed. One student’s summary is representative: “the VR simulation was super interesting and the wide-view videos of the procedures being performed was very helpful” (W25-R044/Q5).

Talk of doing is another matter. The coding distinguished students’ talk about the environment’s content—what there was to watch—from talk about acting in the environment: performing a task, making something, answering a question, choosing a path. Talk of that second kind appears 37 times across the 873 responses, from 33 students, and almost every instance is a request or a complaint rather than a report: students describing not what they did in the virtual hospital but what they wished it had given them to do. One student put the wish in terms of routes: “The VR clinical immersion platform would be better if it was more interactable and had options on which route you want to take rather than just being playback of videos” (W25-R056/Q9). Another’s entire answer to the improvement question was a single request: “Make assignments based on the VR clinical immersion” (S25-R052/Q9). A third wrote: “Maybe add something more interactive like games/checklist/checkpoints etc” (W26-R054/Q9). Requests of this kind came from 53 students, 22% of the class, and they came twice as often from application users (30%) as from students on the YouTube route (15%): the students asking for something to do are, above all, the students who actually used the VLE. Among the 37 instances of action-talk, exactly one reports an action the virtual hospital gave rather than an action it lacked—and it comes from the same student, W25-R044, whose praise stood for the content channel: “The virtual space feels like you are in it, while just a video doesn’t. It makes you more involved and interactive with the space” (W25-R044/Q10).

What that missing act would have interrupted is long, unstructured watching, and students describe that experience in detail. For a student alone in the building—as most were—the world asks three things: find the room, open its player, and watch. Nothing in the world asks anything back. No room poses a question, no procedure ends in a prompt, and the course’s actual work of finding an unmet clinical need in the footage happens outside the world entirely, in notes and documents the platform does not include: “I wish the VR platform would accommodate other open windows. It wasn’t easy to take notes on a Google doc while

³⁹A complete catalogue of every reported friction, ordered from most to least reported, appears in the appendix.

listening to interviews or watching videos on the VR platform, so I resorted to the youtube video afterwards” (W25-R058/Q6). What remains inside the world is watching, and the watching is long. One student found “a lot of space of inactivity. Videos were unnecessarily long” (W25-R025/Q6). Another asked for help with the seeking itself: “I wish there was a bit more guidance in watching the clinical procedures. Some of them are many hours long, and I’m not sure what exactly to be looking for during that time” (S25-R035/Q6). A third met the same difficulty in the course’s own terms: “at times it was hard to figure out a ‘problem’ or ‘unmet need’ that was occurring. I think I was looking at the wrong things in the beginning” (W26-R027/Q9). These students are describing one situation. Each arrived with attention to give and a task in hand—find a need—and the footage gives every minute of the operation the same weight. No moment is flagged as a difficulty; no prompt says *watch how the team works around this*; nothing distinguishes the places where an unmet need shows itself—a workaround, a delay, an awkward instrument—from the hours of routine that surround them. Seeking of the kind the course assigns requires a determinate object—something in the world picked out as what the seeking is *for*—and the dissertation’s term for that object is the *orekton*. The students quoted above report its absence in their own words: interest solicited, seeking underway, nothing marked as the thing sought. In the chain’s terms the finding touches the passage from A_3 to A_4 : the world reliably solicits attention and interest, and the acts it furnishes stop at watching. It comes down to a lack of affordances, of meaningful potentialities for the students to actualize.

The 108 students who watched on YouTube divide into two groups. Sixty-one reported that they could not run the application at all; for them the flight was forced, and the videos were the accommodation the course had built for exactly their case. Their talk is regret as often as complaint: “I wish that I could access the vr setting on macbook because it feels like a great opportunity that I missed but the videos helped to accomadate for me” (W25-R099/Q6). The other forty-seven students reported that they could have used the application and watched on YouTube anyway. For them the world stood open and they declined it. Talk of switching to the videos appears 48 times in the corpus, from 41 students, and it comes from 26% of the YouTube group against 11% of application users—those last being students who entered the world and then left it.

The students who declined the world explain themselves in the language of effort. One, from the group that could have used the application, watched the comparison tilt over time: “the VR videos were cool at first, but the youtube format ended up being simpler to use and watch” (W25-R017/Q9). An application user performed the same calculation from inside the world, room by room: “The effort going into finding a procedure to watch is way more than just clicking on the youtube video, so it would be better if the room had more than just the viewing feed and something to do to make that effort worth it” (W25-R002/Q9). A third student shows how soft the line between the two anatomies really is. On the survey’s yes-or-no item this student answered that they could not use the platform, yet their account of it is a decision, not a defeat: “The software. I ended up just giving up on it and just watching the youtube videos. It’s not worth the effort to even learn how to use it and download it all when I could just pull it up on youtube” (W25-R041/Q6). For some students, that is, “could not” did not name a hard technical wall—the student was a Mac user and could not run the program at all, or the hardware was too weak to run the build—but the point at which they stopped trying: capable in principle of getting the application working, they abandoned the download, the installation, or the troubleshooting when the effort began to exceed what the platform seemed likely to repay. The line between the sixty-one and the forty-seven is therefore softer than the survey’s yes-or-no structure suggests; an unknown share of the excluded students excluded themselves. All three students quoted above are doing the same calculation. Each step the virtual hospital requires—download, install, sign in, launch, walk, find the room, open the player—is a step the YouTube mirror does not, and what waited at the end of those steps was the same footage. One student asked for the steps to be abolished outright: “Maybe not having the place in the game just directly clickable links to open the videos. It was weird having to walk around just to find the right video” (W26-R047/Q9)—the walk that makes the world a place, experienced here as pure overhead between the student and the footage. Friction sharpened the comparison where it appeared: mentions of switching to the videos co-occur with install complaints eleven times, with crashes five times, with latency four. One response carries the whole sequence from occasion to flight: “the platform was extremely slow in terms of accessing the videos and interviews, which eventually became frustrating and led to watching the Youtube videos” (W26-R023/Q9)—the friction named, the frustration concretizing, then the switch. A failed install or a crash was, in case after case, the moment a student switched. But the moment is not the mechanism. The mirror was available before any crash and after every one—always one

click away, always carrying the same footage, never asking for a download, an account, or a walk through a building first. Friction supplied the occasions. What made every occasion sufficient was the comparison that always stood behind it: at any moment of difficulty inside the world, the same footage was already available by an easier route, so the difficulty never had to be weathered—it only had to be noticed.

Watching on YouTube was never against the rules. The course created the mirror channel deliberately, so that no student would be left without the material, and for the sixty-one students whose machines could not run the application—the Mac owners, the weak laptops, the failed installs—it did exactly the job it was built for. But a channel built as an accommodation for students who could not enter the world also stands as an alternative for every student who could. Heidegger’s name for the everyday counter-measure to boredom is passing the time (*Zeitvertreib*)—“a driving away of boredom that drives time on”—the reach for whatever occupation lies nearest when the present one fails to hold us (*Fundamental Concepts* 93).⁴⁰ The flight to video has exactly that structure: not a verdict pronounced against the virtual hospital, but a drift toward the nearest sufficient occupation—and here the drift was institutional, because the course itself supplied the nearer channel and stocked it with the same content. Whenever the virtual hospital offered a student nothing beyond what the videos already offered—the same footage, reached through more steps—the videos won, term after term, at a stable 43–46% of each cohort. One student stated the pattern as a plain fact about the class: “The VR platform is too cumbersome to get working on your computer, most students just resort to the online videos” (S25-R034/Q6).

The dissertation’s frame makes the mechanics of that drift precise. The flight is not a failure of the chain of desire, nor even an unusual run of it: the chain completed through the same mechanisms by which it always completes, and what varied was not the chain but the delivery of its object. What these students desired—the object their coursework required and their interest tracked—was the content itself: the operations, the interviews, the material they needed to view. Nothing bound that object to the virtual hospital; the same footage was available through both distribution platforms. So the chain ran to its end. The footage solicited interest. Between that interest and the act stood a comparison, and the comparison is the work of *phantasia*: in its discursive-deliberative orientational mode, *phantasia* holds a *phantasma* of each path before the mind—the download, the install, the walk through the building on one side; the single click on the other—and runs through routes not yet taken, which is what makes a reasoned weighing of alternatives possible at all. That deliberation issued in the committal *doxa*, the taking-as-true: “simpler to use and watch” (W25-R017/Q9); “not worth the effort” (W25-R041/Q6). In that ratification the emotion concretized—the frustration of the threshold, the impatience with the hassle—giving the judgment its weight. Then came the act: the student watched, and the desire for the content was actualized, on YouTube.

The *Rhetorica*’s sorting of action by its causes names what kind of act this was—and for whom. Aristotle divides all action by whether its origin lies in the agent or outside: chance, nature, and compulsion move a person from without; habit, reasoning, anger, and appetite are the agent’s own (*Rhetorica* I.10, 1368b33–1369a6). The sixty-one students the platform shut out fled by compulsion: the *archē* of their flight was the machine’s refusal, an external wall standing against their own *orexis*—the regret in “a great opportunity that I missed” (W25-R099/Q6) is the sound of a desire overridden from without. The forty-seven who could have entered and did not fled from causes within their own power, and their words name which. What their responses record is comparative reasoning enabled by *phantasia*: these students held both platforms before the mind, measured them by a single standard—the coursework’s need for the content against the effort each platform demanded—and pursued what appeared the greater good.⁴¹ Their retreat was a deliberate one. It rested, moreover, on habit’s floor: YouTube was for every student the often-done thing, the channel whose use had long since become, in Aristotle’s phrase, “virtually natural” (*Rhetorica* I.11, 1370a6–8)—so the familiar route asked nothing of them while the unfamiliar one asked much. One student located the asymmetry in themselves, unprompted: “I preferred the YouTube channel for virtual reality, over the app. I’m not a video game fan, maybe is that” (W26-R007/Q6)—the unfamiliar route is unfamiliar precisely to the student without the gamer’s second nature. The taxonomy also says precisely what the survey’s yes-or-no

⁴⁰Passing the time, Heidegger insists, is never taken up on its own account: it is “a passing the time which lays claim upon us specifically out of and in opposition to a particular boredom” (*Fundamental Concepts* 93). The reach for the nearest occupation is thus itself evidence of the boredom it answers.

⁴¹The operation is the one Aristotle assigns to deliberative *phantasia*: “whether this or that shall be enacted is already a task requiring calculation; and there must be a single standard to measure by, for that is pursued which is greater. It follows that what acts in this way must be able to make a unity out of several images” (*De Anima* III.11, 434a5–10).

structure blurred: the line between the two anatomies of the flight is the line between an *archē* outside the student and an *archē* within. Students like the one who “ended up just giving up on it” answered that they could not use the platform, but their own accounts locate the *archē* of the flight elsewhere. Had the machine truly refused them—a build their hardware could not run, an install that failed no matter how long they worked at it—the cause of their flight would have been compulsion: an origin outside the agent, external circumstance overriding the student’s own desire to enter, the involuntary cause under which the genuinely excluded students fall. What these students describe instead is a stopping-point they chose. The install had not finally failed; they had stopped entering the VLE, on their own deliberation that the additional effort was not worth its return. In the *Rhetorica*’s strict sense their flight was voluntary, the act their own, whatever the survey answer records. The world, in the end, was not refused as a world; it was weighed as a delivery platform for an object it did not exclusively hold, and the effort it asked was too great for a destination reachable in one click. In the chain’s terms the finding touches A₃: the *phantasmata* of both platforms taken up in the discursive-deliberative mode and ratified by a *doxa* in which the emotion concretized—the chain completing as it always completes, deciding not only whether the act would happen, but how, and from an origin that was, for the forty-seven, entirely their own.

Each term the survey put a direct question to the students who used the application: “Did you find the VR clinical immersion platform to elicit a greater degree of immersion, presence, and/or embodiment (the feeling that you are actually present in the operating rooms/clinical environment)?” Of the 123 students who used the application and answered, 87 said yes—70.7%. The majority held in every term taken alone, not merely in the pooled total: 64% of Winter 2025’s application users answered yes, 82% of Spring 2025’s, and 67% of Winter 2026’s. The figure lands with a history behind it. In the pilot, we had asked whether the 3D VR versions elicited a greater degree of presence than the other formats, and 70% of respondents said they did (“Phenomenological Evaluation” 388)—students wearing headsets, standing inside stereoscopic footage. The deployed VLE reproduces the pilot’s number almost exactly, with no headset involved: where the pilot’s question was answered by twenty students in a single quarter, the deployment’s immersion item was answered by 123 application users across three terms, every one of them working through an ordinary monitor by mouse and keyboard. Presence, as the students report it, happened through traditional displays as much as it did through 3D VR.

The yes-answers do not stand alone; the open responses corroborate them in the students’ own words. Nineteen students, unprompted by any yes-or-no item, wrote descriptions of feeling present in the virtual hospital—of being in the space rather than watching it—and thirteen of the nineteen were application users (11% of that group, against 6% of the YouTube group). The fullest of these descriptions comes from the same student, W25-R044, who supplied the corpus’s single account of given action: “The virtual space feels like you are in it, while just a video doesn’t” (W25-R044/Q10). The sentence repays slow reading, because in a student’s plain words it draws the dissertation’s own distinction—and draws it under conditions that isolate what makes the difference. Both routes reach this student through the same hardware: the YouTube videos and the virtual hospital alike arrive on an ordinary computer display. Whatever separates being-in it from watching it therefore cannot be the screen; it is the design—how the same footage is framed and delivered to the student who watches. Nor is traditional video barred from disclosing a world: a film seen on that same monitor can take its viewer into its world, and *phantasia* is active on both routes. The difference lies in what the design gives *phantasia* to work with. The YouTube route delivers the footage as one video among videos—a page, a player, a list of what to watch next—content framed as content, the surround reasserted at every edge of the frame. The VLE delivers the same footage as the interior of a place: to reach it the student has walked a building, found a room, and stood before its screen, and the hospital continues around the footage while it plays. What the student’s statement registers is the difference between those two relations—a world disclosed as a place dwelt in, against a scene surveyed from without. That difference is what the dissertation names world-disclosedness: the student taken into the world acted in, and the world taken into the student’s awareness. On a flat screen that disclosure is *phantasia*’s victory in either case. A monitor gives the eyes a bounded rectangle inside a room that remains fully present around it, and it is *phantasia* that subtracts the surround, supplies what the screen withholds, and encloses the student in the world the screen only samples. The designs differ in how much they help: the virtual hospital hands *phantasia* a surrounding place already built—an inside to be-in—where the video page hands it a frame to be overcome. The coded responses register that difference in help directly: of the nineteen students who described being-in, thirteen came from the 123 application users and six from the 108 on the YouTube route—students given the built

world reported being-in at nearly twice the rate of students given the framed video, though both watched the same footage on the same kind of screen. A second description extends the being-in past place into role: “I loved the VR videos; I feel that it actually made me feel like an engineer trying to find and solve problems that real doctors are facing” (W26-R007/Q5). The being-in here is vocational as much as spatial: the student reports becoming the engineer the course wants them to be, the seeing already oriented toward the problems real doctors are facing. The same student, in another answer, preferred the YouTube route, tracing the preference to being “not a video game fan” (W26-R007/Q6)—a reminder that the presence the footage occasions and the route a student settles on are separate verdicts, delivered here by one voice.

This disclosure is also the first face of what the dissertation calls rhetorical incorporation. The term names the union a designed world achieves with its inhabitant, and the actualization chain states the condition on which that union is fully achieved: the world solicits perception (A_1), *phantasia* composes the charged image of a surrounding place (A_2), the student’s committed judgment takes the world as worth engaging (A_3), and the union is fully actual only when that judgment passes into an act the student performs in the world (A_4). It is worth working through what that means for these students. Feeling present is not the completion of rhetorical incorporation; it is the world’s half of an exchange. The chain is understood backward from its end—from the desired object whose pursuit completes it in an act—so a world that has disclosed itself has, so far, only made acting possible: it has given the student a place in which an act could happen. Rhetorical incorporation is finished when the student’s own act, done in and with the world, ratifies the disclosure—when being-in becomes acting-in. A world that discloses but does not task therefore leaves the exchange half-done. The student stands in a place that asks nothing of them; the disclosure that invites completion remains an invitation; rhetorical incorporation is begun and then suspended at the very threshold where it would become actual. That is the precise sense in which such a world still owes its students an act: the debt is written into the chain’s own teleology, where disclosure is for action as means are for ends. Other students put the being-in more modestly—“Being able to get somewhat first hand observations” (W25-R089/Q5); “I found the immersive VR helpful” (W26-R028/Q5)—but the relation they report is the same. One more names what the being-in held off: “It was different for each place and being able to make it seem like we were in the procedure and watching made it more immersive rather than boring” (S25-R028/Q5)—immersion, in this student’s own accounting, standing where boredom would otherwise stand.

Nine students asked the course to provide VR headsets: “This class should provide students with VR headsets. Even if there were a technology fee, it would be more beneficial for learning” (W26-R014/Q9); “I feel like we should have some sort of cheap VR set because the VR experience feels off when seeing it in the laptop screen” (W25-R020/Q9).⁴² What these nine notice is the division of labor just described. Every medium supplies some share of what the senses would receive in the non-virtual scene, and *phantasia* must furnish the rest: a headset supplies much and asks little of *phantasia*; a flat screen supplies least and asks most. The student who writes that the experience “feels off... in the laptop screen” is reporting how large *phantasia*’s share becomes on the desktop route—feeling, in the moment of viewing, how much of the world it falls to their own *phantasia* to hold up. What the 70.7% then says is that the share was held. Seven in ten application users, on the route where the medium supplies least and *phantasia* must supply most, reported the feeling of actually being present in the operating room.

Set the pilot’s number and the deployment’s side by side, and they sharpen into a finding about where immersion comes from. The pilot’s students wore stereoscopic headsets and stood inside the footage; the deployment’s students sat before flat monitors; both cohorts reported presence at 70%. If the stereoscopic viewpoint were what carried immersion, the number should have fallen when the headsets disappeared—and it did not fall. What the two cohorts shared was not display hardware but a designed, explorable world: the virtual hospital, entered and navigated, on whichever display. The likeliest reading of the matched numbers is therefore that the immersive power *phantasia* conveys derives not from the stereoscopic viewpoint but from the inclusion of the footage in an explorable world. Calleja’s account of incorporation, built from screen games—players at monitors, no headset anywhere—predicts exactly this. He defines incorporation as “the absorption of a virtual environment into consciousness, yielding a sense of habitation, which is supported

⁴²The request inverts the pilot’s most curious result. In the pilot, we housed several Meta Quest 2 headsets in the school’s library, and “not a single student went there to check one out”; as one student phrased it, obtaining a headset “was a hassle” (“Phenomenological Evaluation” 389). Three years on, students ask for headsets to be handed to them. The desire for the richer medium is real in both cohorts; what governs it is the same convenience gradient that routes everything else in this deployment—intensity is wanted, but only at zero fetching cost.

by the systemically upheld embodiment of the player in a single location, as represented by the avatar” (*In-Game* 169): habitation grows from emplacement in a navigable world, not from any property of the display. Of the six dimensions of involvement his model blends into incorporation, the cornerstone pair without which it cannot occur is the *spatial* and the *kinesthetic*—the internalized space and the learned movement through it (*In-Game* 170)—and that pair is precisely what the VLE adds to the footage and YouTube does not. Behind the mechanism stands a principle of Merleau-Ponty’s: “What counts for the orientation of the spectacle is not my body, such as it in fact exists, as a thing in objective space, but rather my body as a system of possible actions, a virtual body whose phenomenal ‘place’ is defined by its task and by its situation. My body is wherever it has something to do” (*Phenomenology of Perception* 260).⁴³ The explorable hospital gives the student’s virtual body somewhere to be—a place constituted by situation and task, a room to find and a player to open—on whatever display the world arrives through. In the chain’s terms the finding sits at the *phantastic* motion that renders A₂: from the thin *aisthēmata* a monitor delivers, *phantasia* still composes the charged *phantasma* of a surrounding world—a world taken-as dwelt in—and it composes it from the explorable place the design supplies, not from the apparatus on the head.

***** FIGURE PENDING *****

Insert screenshot: multiple students meeting inside the VLE (group co-watching scene).
Candidate source: the 2024 group-session capture.

Of everything the virtual hospital offers, one affordance has no YouTube counterpart at all: the VLE is built for watching together. Students hear the classmates near them in the space, can share their screens there, and every student belongs to a procedure-paired group from the term’s first week; the designed use of the environment is that the group convenes inside the world and watches as a group, and that multiplayer functionality is clearly stated and encouraged in the course materials. What the survey can say about that design is limited in a way the other findings are not. Unfortunately, we did not think to ask the students whether or not their group convened in the world, so no rate of use statistics are available. Given the group based nature of this class, we falsely assumed students would organize meetings within the VLE of their own volition, and that false assumption led the course and survey designers to ignore crucial questions to be asked and course designs that could have mitigated such. What the 873 responses hold instead are incidental mentions—remarks students happened to volunteer about watching together or about being alone—and counts built from such mentions can only set minimums. That one student described co-watching means at least one group convened; it cannot mean only one did, because students who met and never mentioned it are invisible to the survey. Each count in what follows is therefore a floor—the minimum established by those who happened to speak—and never a rate. After reviewing the relevant 115 candidate responses by what each actually reports and sorting them, they fall into four different kinds: reports that the group meeting never happened; a report that it did; requests that the course structure the meeting; and requests for a co-use capability that, in nearly every case, the platform already possessed.

The best-attested kind is absence. Eleven students report directly that the meeting did not happen for them. One logged in and found the building empty: “I found that when I did log in, I was usually alone in the platform so it didn’t really make me feel more connected with my peers” (W25-R038/Q10). One’s group met, but elsewhere: “The virtual space to meet up for teamwork was very smart, but my group utilized discord a lot” (S25-R072/Q10). One knew the capability existed and never used it: “Meeting up with other people in the virtual space is an interesting idea but because I didn’t actually experience it all

⁴³Wendt reaches this same passage from the design side: for him it carries direct consequences for the design of experience, which must orient itself to what users are there to do—their goals, conscious and non-conscious—within the multiple stable uses any technology admits (*Design for Dasein* 126). The convergence is the point: what the phenomenologist states of the body, the design theorist prescribes for the world built around it—a world is designed well when it gives the virtual body something to do. A third tradition arrives at the same structure independently. Chalmers, with no reference to Merleau-Ponty, builds his defense of virtual realism on the same figure: “If I pick up a virtual coin in Second Life, I really use my virtual body to take possession of a virtual coin. There is nothing fictional about this” (“The Virtual and the Real” 317); “virtual actions are plausibly real actions (albeit with a virtual body), so that when one performs virtual actions, one really is doing something” (339); and a user’s sense of ownership over a virtual body “need not always be an illusion” (333). Phenomenology states the principle—the body is where its tasks are; design prescribes it—build worlds that give the virtual body something to do; analytic metaphysics certifies its object—what the virtual body does is really done. For the survey’s 70.7% the triangulation matters: the students’ reports of being present in the operating room need not be read as a flattering illusion, since they describe a virtual body genuinely located by task and situation. It sharpens the deployment’s deficit in the same stroke, for a virtual body whose acts are real still needs acts to perform.

I can say is it seems cool in theory” (W25-R014/Q10). A fourth asked for “more of a community since it just seemed isolating” (W25-R089/Q6); a sentence that reads as a complaint about the design until it is set beside the platform facts: the community infrastructure existed, and what this student experienced was a groupmate-less default, not an absence in the world. The isolation reported in these responses is real. Its cause, in every case the responses let us see, was that the assigned group never convened where the course design intended and which was heavily encouraged.

Against these eleven stands one confirmation that the designed use happened. In all 873 responses, exactly one student reports co-watching as an experienced feature of the platform: “It is more effective that it does put you into a VR clinical environment and a cool interactive space. The main pro to the platform is you can watch with others” (W26-R037/Q10). No second response confirms it. The rest of the WE talk is wishing, and the wishes split in two tellingly different ways. Four students ask for the meeting to be structured—not for the capability, but for someone to carry the organizing: “Incorporating a required group meetup within the VR simulation could incentivize more interactive learning” (W26-R052/Q9); a class sheet “where times are set by each group to enter the VR program” (W25-R060/Q10); “remote teamwork VR space weekly meetings, to ensure accountability” (S25-R040/Q10); an instructor logged in at the same time as the students, explaining as they watch. Thirteen more ask for co-use capability in wording that implies it does not exist—and for twelve of the thirteen, the request is for something the platform already has. “I think if there was a way to interact with other users who are online at the same time that would add to it” (W25-R071/Q9); “perhaps the ability to meet up with groups in a virtual reality setting” (W26-R022/Q10)—the platform’s existing capability, requested as an innovation. The gap the survey shows is therefore double: activation—groups that could have met and did not—and awareness—students who never learned the meeting was possible at all.

Read together, the counts show why the meeting so rarely happened. Every other use of the platform asks one student to act alone: download, install, launch, walk, click play. Watching together is the single use that requires two or more students to act at the same time—agree on a time, log in together, find the same room—a demand none of the solo uses make. The counts reflect that demand directly: one student’s responses confirm a group that met in the world; eleven report groups that never did; and seventeen ask that the organizing be done for them, by the course, the instructor, or a posted schedule. The same pull of convenience that sent students to YouTube worked against the meeting as well, and it worked hardest against the one affordance the YouTube platform cannot reproduce: watching together, in one place, at one time. In the dissertation’s terms, what stood ready on this route was co-disclosedness—the shared counterpart of world-disclosedness. The students who described feeling present in the virtual hospital were each alone in it: taken into the world, but singly, one student before one monitor. Co-disclosedness is that same being-taken-into-the-world had together with others who are present in it at the same time, and it is the half of the disclosure the platform’s social architecture was built to make possible. The survey states its standing directly: every student had it available by design, and one student’s responses confirm that it was ever had.

The deployment’s last finding lies in the difference between its terms, and it must be read modestly: the three cohorts differ in size and possibly in makeup, so what follows is a trend to be described, not a test that was passed. The trend begins with the frictions falling. Crash complaints numbered 21 in Winter 2025 and 4 in each term after; hardware complaints fell from 8 to 4 to 1. By its third term, the platform largely worked. What rose as the frictions fell was a different kind of talk—talk that questioned the platform’s value rather than its function. A keyword search, run with the same fixed list of search terms across all 873 responses so that every term’s cohort was screened identically, finds 16 responses of this kind: five, two, and nine across the three terms, which is 1.3%, 0.7%, and then 4.2% of each term’s responses—the largest share by far in the final and smallest cohort.⁴⁴ The late voices are not reporting failure. “I think the youtube videos were enough to fully grasp the experience” (W26-R033/Q9). The register is not confined to the final term; its fullest argument came in the middle one: “the vr learning platform wasn’t accessible, and i think if youtube videos do the job, they do the job. some things don’t need to be improved on; take the paperclip

⁴⁴The counts come from a keyword sweep applied uniformly across all three terms rather than from the coded label for this theme, because that label was invented mid-run by one batch’s coders and its all-Winter-2026 distribution reflects coder vocabulary, not the terms. The sweep found 16 responses (two further keyword matches were inspected and discounted as false positives). The theme itself is not an artifact: independent blind coding passes rediscovered it in Winter 2025 under other names. One caveat stands—the sweep’s vocabulary is partly seeded from Winter 2026 phrasings, so the exact size of the rise is soft, though its direction is robust.

for example” (S25-R039/Q6)—sufficiency argued by analogy to a perfected design. “Less dependency on the VR aspect altogether” (W26-R056/Q9)—that is one student’s entire answer to the improvement question. A third goes past sufficiency to a failure of imagination about any use at all: “I do not think this one should ever be mandated in the future. It should be an extra option. I cannot imagine a time when a VR space would be more helpful” (W26-R055/Q10). Traditional learning, a fourth wrote, is “the least gimmicky” way (W26-R022/Q10). One voice adds a complaint of another kind entirely—not that the world is unnecessary but that it fails to look serious: “The VR clinical immersion lobby is modelled poorly and it reminds me of a children’s game. I didn’t find it to be very academic? The initial impression really turned me off of using it and the actual videos were disorienting” (W26-R024/Q9)—the game-like look that makes the world enterable, undermining at first sight its authority as a clinic. The complaint has changed kind: in the first term students told us the world did not work; by the third, with the world working, they asked why it should exist.

A small set of responses sharpens that question: in one and the same answer, these students praise the platform and report that something about it felt empty. Thirteen responses carry the pattern. Two of them—submitted in different terms, under different respondent codes—are word-for-word identical, so the analysis counts that text once, which leaves twelve distinct responses that praise the platform while finding it hollow. Read closely, most of them resolve into ordinary complaints, because the hollowness has a named culprit. “The actual idea of having the VR clinic is amazing, the videos and images that come with it not so much” (S25-R022/Q9)—the culprit is the footage. “The VR is a cool idea but the quality is so low I don’t gain from the atmosphere and I don’t feel immersed” (W25-R014/Q9)—the culprit is the quality.⁴⁵ Complaints of this shape are repairable in principle: fix the named thing and the praise stands alone. Two responses, though, name nothing. “The virtual reality environment, while it allows for more immersion, wasn’t the highlight of the course, and could be amended for something else” (W26-R016/Q6). “It is a cool concept, but I feel like at the moment it is not doing much to help regarding this class or the assignment” (S25-R009/Q10). Nothing is broken in these responses; something is missing, and the students cannot say what. The sharpest case is a pair of answers by one student—the same student whose response gave the survey its one confirmed co-watching. Asked about educational use, this student praised the platform in the survey’s fullest presence language: it “does put you into a VR clinical environment,” and “you can watch with others” (W26-R037/Q10). Asked what to improve, the same student wrote that the platform “is really laggy at times and is not significantly more immersive than a YouTube video” (W26-R037/Q9). One voice, both sides: the world entered and shared, and the world flattened to a comparison it barely survives.

What this register can testify to has a stated ceiling. In the dissertation’s account of boredom, which will be analyzed in the next section, there is a form in which nothing determinate is wrong—the occupation is intact, even pleasant, while the whole of it quietly empties: an occupied surface over a hollow depth.⁴⁶ That form is detectable only where two channels of evidence can disagree about the same experience, as the laboratory’s physiological record can disagree with its behavioral one. A survey holds one channel: what a student noticed, and chose to write, about their own experience. The laboratory could set a brain measure against a gaze measure and let the two contradict each other; a survey response has nothing set against it that could contradict it. The two responses quoted above that praise the platform while naming no culprit—W26-R016’s and S25-R009’s—therefore show the same shape as this form of boredom: commendation of an occupation that works, standing over an emptiness the student cannot name. The shape matches;

⁴⁵The perceived lack of quality has two main sources. First, the program does not detect and automatically apply the native resolution of the student’s laptop; the student must set it manually in the graphics settings menu. Students are provided with onboarding videos that state this clearly and walk through how to adjust the settings to optimize quality and performance for their specific hardware, but it seems many never actually do so. Second, the quality can suffer from limited internet bandwidth. Each operation’s footage comprises two or three simultaneous video streams: the 180- or 360-degree stereoscopic view of the operating room, the physician’s point-of-view video, and, in some cases, recordings from surgical equipment such as the systems used to conduct eye surgeries. The playback system adapts to each student’s bandwidth, serving lower-resolution footage when bandwidth is limited, but playing the streams at 4K resolution requires well over 100 Mb/s. The same videos uploaded to YouTube must be baked together into a single stream, so only one video is technically playing; and YouTube’s compression and streaming systems, being substantially more advanced than those I was capable of developing, will always deliver better quality for the same bandwidth. Where my system shines is in the ability to relocate the point-of-view videos while they play, or to hide them entirely, letting students place them where they see fit for a better viewing experience or clear them away to focus on the panoramic view of the operating room.

⁴⁶The form is the second of the three Heidegger distinguishes in *The Fundamental Concepts of Metaphysics*: becoming bored with an occupation, as against being bored by a determinate thing (*Fundamental Concepts* §§24–28). The full apparatus—the three forms, their signatures, and what each requires of evidence—is set out in the next section.

the survey cannot confirm what lies beneath it. This section reads them as echoes of the form, never as detections. What the survey can show directly is where the arc of complaint ends. One further fact belongs to that ending: the student who called the platform gimmicky went on, in the same response, to propose its remedy—“perhaps the ability to meet up with groups in a virtual reality setting” (W26-R022/Q10)—the platform’s existing, unactivated capability. The survey’s most skeptical voice, answering a question about where virtual environments could serve learning at all, proposes the one affordance this platform offers that YouTube cannot: meeting inside the world, together. The arc ends at a question no repair can answer: not whether the world works, but what it is for—the justification the world itself must supply.

5 The Analysis: Boredom, Justification, and the WE

A world built for recreation never has to say why it should be entered. Its player arrives already seeking something—an evening’s diversion, a story, a game worth playing—and the world reserves its arts of solicitation for the encounters the player did not seek. A course world begins from the opposite condition. The student arrives because the course requires it, and so the solicitation that recruits them cannot come from inside the world at all. It comes earlier, and it comes in language. From the outset we presented the environment as the place where a procedure group could watch its assigned footage together and talk while watching—group analysis conducted inside a shared clinical space, a way of working the course offered nowhere else.⁴⁷ That announcement is the deployment’s act of solicitation. It names the reason to enter this world rather than the YouTube platform that carries the same footage. What remains for the rendered world itself is the quieter half of the work: not arresting attention but keeping it, across the weeks of a term, in a world the student was assigned rather than drawn to.

Judged by what it set out to sustain, that quieter campaign succeeded. Three of every five students wrote something affirming the footage’s interest, the largest positive register in the collected data; presence language came from those who entered the world; attention, once gathered, was held. The deepest complaint in the data is not that the world failed to draw anyone. The campaign goes silent at exactly one point. Interest that has been brought to commitment wants something to complete itself in, and completing it would require the world to stop soliciting and begin tasking—to station an *orekton*, a determinate object for desire and action: something to find, perform, or submit from inside the hospital itself. The world contains no such station. There is no procedure to attempt in its rooms, no needs-finding to capture where the footage plays, no deliverable the world can receive. In the chain’s terms, the deployed world reliably carries a student to committed interest at A_3 ; a crossing to A_4 always follows, because the chain runs always, and the question is which potentiality that crossing actualizes. The only potentiality this world furnishes it is watching. The hail itself made this silence hard to see. What we announced was a place for the group’s analysis, not an analysis the world would conduct; the task always lived on the course’s side, in deliverables the world never housed. A pitch can succeed—students entered—while the world it recruited for gives the interest it sustains nothing beyond watching to complete itself in. A world that can only be watched has a boredom half-life.

None of this makes boredom a property of the footage. The same virtual hospital drew the survey’s fullest presence praise and its flattest comparison, and its sharpest witness is a single student—the survey’s one confirmed co-watcher—who wrote in one answer that the platform “does put you into a VR clinical environment” where “you can watch with others” (W26-R037/Q10), and in the next that it “is not significantly more immersive than a YouTube video” (W26-R037/Q9). Nothing about the hospital changed between those two answers; the relation did. What holds for the footage holds for the route as well. A student who tires of a long surgical recording will tire of it in the virtual hospital and in a browser tab alike; I take the two routes to differ in how soon that tiring arrives and in how far it goes, not in whether it comes. Boredom, then, cannot be what separates the two routes. Whatever it diagnoses, it diagnoses the watching itself—the situation of being given something to see and nothing beyond watching to do, which the deployment and the YouTube platform stage alike. Saying what that boredom is, rather than merely that it occurred, requires distinctions of structure rather than degree. In the dissertation’s account, boredom takes three forms; the forms differ in what they attach to, and they differ just as sharply in what evidence can show them.

⁴⁷The platform’s published description names the affordance directly: the environment “supported real-time voice chat and screen sharing, enabling collaborative team viewing if students wanted to meet in groups within the virtual environment” (“Scalable Virtual Reality” 4).

Heidegger's lecture course of 1929–30, published as *The Fundamental Concepts of Metaphysics*, gives boredom one of the most sustained, philosophical, and structural analyses it has received, and the analysis begins from a definition small enough to carry: what bores is “that which holds us in limbo and yet leaves us empty” (*Fundamental Concepts* 87). Boredom, on this account, is always two things at once. Something holds us: we cannot simply leave, because the situation keeps us in it. At the same time, what holds us offers nothing, so that we are left empty while being held. Against this double grip we deploy a counter-move Heidegger calls passing the time (*Zeitvertreib*): occupations taken up not for their own sake but to keep the boredom off. The three forms he distinguishes are three depths to which this machinery can reach. Setting them out requires one term of method. Call a channel any independent way an experience shows itself in evidence: what a person says or writes about their experience is one channel; what their visible behavior shows is a second; what an instrument reads off the body is a third. The forms of boredom differ in how many channels it takes to see them, and for the collected data that difference matters as much as the forms themselves.

In the first and shallowest form, we become bored *by* something. A definite thing is the culprit—a book, a lecture, a wait—and we could name it if asked, because the boredom stays attached to its object. Heidegger's scene is a missed connection at “the tasteless station of some lonely minor railway”: “It is four hours until the next train arrives” (93). The wait holds us—we cannot leave without losing the train—and the station offers nothing, and so we fight, in the open: we read the timetable, count trees, draw figures in the sand, and “catch ourselves looking at our watch yet again” (93). The repeated glance at the watch is what Heidegger calls a “helpless gesture” (97): it shows that the passing of time is failing against the wait. Everything about this form is visible, and that is its evidential mark. Because the culprit can be named and the fight can be seen, a single channel suffices to attest it. A student who writes “[t]he VR is a cool idea but the quality is so low I don't gain from the atmosphere and I don't feel immersed” (W25-R014/Q9) has named the culprit inside the complaint itself; no second instrument is needed to confirm what the sentence already carries. The collected data holds this form's whole anatomy in that one channel. The culprits are named: the footage's quality, its length, the lag. The occasions are named: the crash at startup, the buffering pause, the room that could not be found. The counter-move is named too, because in this deployment the readiest way to pass the time spent waiting on a struggling world was to leave it for the same footage elsewhere: “the youtube format ended up being simpler to use and watch” (W25-R017/Q9). Culprit, fight, and exit, all written down in so many words.

The second form is stranger: we are bored *with* something, and no culprit can be found. Heidegger's scene is a dinner party. The food is good, the talk is lively, nothing drags; asked at the table, we would say, sincerely, that we are having a fine evening, and looking back on it we can confirm that “[t]here is nothing at all to be found that might have been boring” (110). Only afterward, at home, does the recognition arrive: “I was bored after all this evening” (110). Nothing was wrong with the evening, and that is the finding itself. Where the first form's emptiness comes from a thing that refuses us, here “what is boring does not come from outside: it arises from out of Dasein itself” (128): the emptiness forms in us, underneath a surface that remains genuinely, pleasantly occupied. The form has a scene any student would recognize: an evening of episode after episode, each one actually watched, none of them bad, which leaves one hollowed rather than restored—nothing to blame, and yet the whole of it empty. This structure—an occupied surface over a hollow depth—is what makes the second form hard to evidence. The surface, asked, would deny boredom and mean it; the depth cannot be asked. One channel can therefore never show this form: a written answer reports the surface, and the surface really is occupied. It becomes visible only where two channels can disagree about one and the same experience, as in the laboratory arm of this study, where a brain measure can read boring-like on the very footage a gaze measure reads as engaged. A course survey holds one channel. It can catch, at most, the form's echo—praise of an occupation that works, standing over an emptiness the student cannot name, the shape of the survey's two culprit-less responses⁴⁸—and that is why the deployment's divergence

⁴⁸“The virtual reality environment, while it allows for more immersion, wasn't the highlight of the course, and could be amended for something else” (W26-R016/Q6); “It is a cool concept, but I feel like at the moment it is not doing much to help regarding this class or the assignment” (S25-R009/Q10). The insight the pair provides is structural. Each response commends the platform—the first even grants it the immersion it promises—and each reports an emptiness without naming anything broken. The hollowness attaches to no defect but to the whole (“wasn't the highlight,” “not doing much to help”), and in the second response it attaches to what the environment is *for*: its help toward the class and the assignment. That is the second form's silhouette in survey prose: the occupied surface speaks well of the occupation while the emptiness shows through only as a want the student cannot locate. With one channel it remains a silhouette; nothing in either response can confirm what

register was read as echoes, never as detections.

The third form is the rarest and deepest, and it strips away even the person who is bored. Its name is impersonal: “it is boring for one” (135)—not for me, with my roles and plans and reasons, but for one, for anyone at all. No thing is the culprit here and no occasion is required, because what goes flat is not this or that offering but beings as a whole: everything at once, oneself included, becomes indifferent. Heidegger’s example is chosen for its very lack of incident: it is boring for one “to walk through the streets of a large city on a Sunday afternoon” (135). The example works because nothing in it is boring. The shops are shut, but one wanted nothing from them; there is no wait, no missed train, no evening to see through; the afternoon lacks nothing in particular—and precisely there, with nothing particular refusing us, everything refuses at once. A student’s version is near to hand: a Sunday on which the assignment, the shows, the group chat, the whole of one’s own room stand available, and none of it—not even oneself—holds any pull, and looking for something to do feels pointless before the looking starts. That last detail is the form’s signature. In the first form we fight the boredom; in the second we fail to notice it; here we do not even attempt the counter-move, because “all passing the time is powerless against this boredom” (135)—one is instead “compelled to listen” to what it discloses (136). Its evidence is accordingly stillness and time: a comportment that has stopped searching for occupation, and a while that “becomes long” (152), felt duration stretching far past the clock. A retrospective course survey can capture neither, and this analysis claims no instance of the form. Where anything of its shape appears in this study’s evidence—in the laboratory’s stillest episodes and dilated felt-time reports—it is read under an announced caution, as analog rather than instance.

Read with this apparatus, the collected data sorts almost entirely into the first form—honestly so, since a survey is a single channel and the first form is the one a single channel can hold. What the apparatus adds is the structure beneath the sorted complaints. Heidegger names the two moments that constitute every form: being left empty (*Leergelassenheit*) and being held in limbo (*Hingehaltenheit*). His most exact observation about the first is that emptiness is not absence: “To leave empty means to be something at hand that offers nothing” (103); things “leave us empty precisely because they are at hand” (102). The virtual hospital is at hand in every sense that matters—rendered, explorable, stocked with footage, praised in the survey’s fullest presence language—and precisely as such it leaves empty, because what it puts at hand offers everything to see and nothing to complete. The station’s refusal even has an address: “The station at hand refuses itself to us as a station and leaves us empty because the train that belongs to it has not yet arrived” (103). The virtual hospital refuses itself as a clinic in exactly this way. A clinic is a place where procedures are performed and needs are found; this one holds only their images, and its train never arrives—no task pulls in to make the place what its rooms announce. One student stated the resulting emptiness with precision: the procedures are “many hours long, and I’m not sure what exactly to be looking for during that time” (S25-R035/Q6)—at hand, and offering nothing determinate to look *for*. The second moment completes the situation. At the station we are held by an interval with a terminus: four hours, then the train. The deployment’s students are held by mandate, with no terminus inside the world itself—assigned footage, required hours, a term’s duration, and, in the small, the buffering pauses that hold a viewer between intention and image. The deployment thus realizes both of Heidegger’s moments at once: its students are held—by mandate, by assigned footage, by the buffering pauses—before a world whose acts stop at watching. The two moments meet at exactly one place, and the location is the evaluation: everything upstream of that point works, so the emptiness students report belongs to the single link the deployment left unbuilt.

The diagnosis follows, and it is a diagnosis of a situation, not of a platform. Boredom, as the collected data presents it, marks a mis-attunement among three terms: a learner who arrives under mandate, with no leisure motive for the world to ride on; an ecology that does its sustaining work, as the data shows it doing; and a task that has no address inside the world. The interest the world sustains is not a contemplative looking that completes itself in every moment of its exercise. It is interest underway toward doing—the students’ requests for assignments, for interaction, for something to find make its direction explicit—and motion of that kind is, in Aristotle’s terms, *energeia atelēs*: incomplete actuality, actuality whose end lies beyond it (*Physics* III.1, 201a10). A world that sustains the motion while housing no terminus does not merely disappoint the interest; it holds it, unfinished, for as long as the mandate runs. The deployment’s shortfall, then, was never bandwidth—presence survived the monitor—but completion. This is the precise sense in which the deployed world’s emptiness is regional. Nothing upstream is withheld: the world is

stands beneath it.

entered, rendered, watched, even dwelt in; the being-left-empty is stationed at the chain's last link, where soliciting would have to become tasking and the world falls silent. The boredom the data attests is therefore not a verdict on the footage, and not a difference between the two routes; it is the felt shape of the watching situation both routes stage, and, unlike a crash or a buffer, it names nothing a repair could answer. The arc of the data shows as much: the frictions fell term by term, and what rose in their place was the question of what the world is for. An emptiness stationed where action should begin asks for one thing only—a for-the-sake-of-which, a reason this world, rather than a video, is where the work of the course happens.

A rendered world, if it is to do anything to the person who enters it, must satisfy a double condition: *veri(dis)similitude*. It must be verisimilar enough—familiar enough in its look, its workings, its ways of being moved through—that a person can enter it and dwell in it at all; and it must be dissimilar enough—different enough from what everyday life already offers—that the entering changes something. A world too familiar struggles to move; a world too strange opposes being entered. For the deployed virtual hospital, the second half of this condition is measured against an unusual standard. Every procedure the hospital holds also plays on the YouTube platform, one click away, free of installers, sign-ins, and crashes. A student deciding whether to enter the world is therefore not weighing it against everyday reality; they are weighing it against the video of itself. *Veri(dis)similitude* accordingly takes a second and sharper form. Ordinarily a rendered world earns its dissimilarity against everyday life, out of possibilities everyday life does not hold. The deployed hospital cannot earn its difference there, because its substance—the procedures—exists identically outside it. Subtract the footage from both routes and compare what remains. On the YouTube platform the remainder is a familiar player and its conveniences: the progress bar, the subtitles, the speed control. In the virtual hospital the remainder is everything the footage plays inside of: the walk down the corridor, the room found and entered, the screen on its wall, and the possibility of another student standing at the same screen. The world justifies its existence with that remainder or not at all. The demand is not unique to virtual worlds. A film that can be streamed at home identically must give a viewer a reason to go to the theater, and the theater's answer has never been the film, which is the same in both places; the answer is the hall, the ecology, the environment, the occasion, the others in the seats, and the ambience. So with the deployed world: to justify its existence it must be different enough from the video of itself, and the difference can be made only of what surrounds the footage—never of the footage, which both routes share.

As the collected data demonstrates, the cohort arrived at this same condition from inside it. As the platform stabilized and the complaints about function fell away, the talk that rose in their place questioned existence rather than function: “I think the youtube videos were enough to fully grasp the experience” (W26-R033/Q9). A question of that kind, put to a world whose footage plays elsewhere, is the difference-condition stated in a student's words. One student stated not only the question but the condition itself, in full: “There is no difference between virtually moving through a museum and seeing photos with text underneath for the purposes of learning”; a virtual environment earns its place only if it “can provide an experience that is unable to be achieved without it, and allows for learning that is done at one's own pace and with elements that rely on the immersion potential and interactivity”—otherwise “there is no purpose in using it over other alternatives” (W26-R059/Q10). This is the student the sign-in gate stopped at the threshold; the voice the door turned away is also the survey's clearest theorist of what would justify walking through it. The question was not unanswered. It had been answered before the first term began, in the announcement that recruited the class: the difference was company—a procedure group watching together, inside the world, talking while it watched. What the data then shows is that difference going largely unattended. Students watched alone, found what a world minus its difference amounts to, and asked why it existed. The survey's most skeptical voice—the student who judged traditional learning “the least gimmicky” way—proposed, in the same response, “perhaps the ability to meet up with groups in a virtual reality setting” (W26-R022/Q10): the difference itself, proposed back to the course that had announced it. The question the deployment ends on is an answer it began with, delivered and never redeemed.

What could the difference be made of? Not the commodity features of delivery, which the students' own comparisons settle. The YouTube platform streams the footage at higher quality for the same bandwidth, carries subtitles, and changes playback speed at will; one student weighed the trade directly, having “gained more from the youtube videos because I could watch in much higher quality which made up for the slight reduction in ‘immersion’” (W25-R014/Q6). On every feature a video player can have, the player built for videos wins. Three materials remain that no player has: a task, a place, and company. A task the deployed world did not contain; that absence is the diagnosis already made. Place it did contain, and the place

deserves its full weight. This class has never had a room. Its students enroll across nine university campuses whose calendars do not even align; there is no lecture hall, no laboratory section, no building, no campus they share. The virtual hospital is the only room the class has ever had—not a supplement to a classroom but the sole site at which this cohort could be co-present in any sense at all. The place also did quiet cognitive work for at least one student: “I found the online course format very useful—especially the VR classroom space. I thought it was really cool to navigate, and helped compartmentalize individual findings” (S25-R048/Q5)—the building’s rooms functioning as an organizer for the student’s own analysis, a use no video list affords. Yet the collected data shows place alone not sufficing. Presence was had on the desktop route and attested plainly, and the flight to the YouTube platform happened anyway; a place entered alone, with nothing to do in it, reduces in use toward the very thing it competes with—a scene to survey, reached the harder way.

Two of the three materials are not wholly beyond the YouTube platform’s reach. Consider the task first, in the course’s own terms.⁴⁹ The syllabus grades three instruments: weekly check-in quizzes keyed to each week’s lecture, an individual innovation notebook in which unmet needs are recorded and evaluated, and a team final report. The weekly schedule sends every group into the virtual environment—making a login and beginning to explore in the second week, then working through the procedure rooms and physician interviews in the weeks that follow—but each graded deliverable is completed and submitted outside the world, on the course page. A check-in quiz answers to the lecture, not the footage; the innovation notebook is written wherever the student writes; the final report is drafted with the team, in a document. Work of that kind runs as well under a YouTube link as inside the virtual hospital, because the course administers it and the student completes it outside the footage—no part of it needs the world. A world could go further: it could make the finding of needs something done in its rooms, captured where the footage plays. The deployment built no such task, and until one is built, the task-material does not separate the two ways of watching. The place can be partly imitated as well. The footage itself is 360-degree film, and the deployment’s own fallback delivered the same videos “with interactive 360-degree panning” (“Scalable Virtual Reality” 4): a student in a browser can drag the view around the operating room and look wherever they choose. What the browser cannot give is everything around that view—the corridor, the doorway, a position in the room—so the pannable video is a thin, surveyable shadow of the place; but it is a shadow of it. The meeting admits no imitation at all. Watching together in a shared rendered room, talking while the procedure plays, is not a feature a video player lacks in degree; it is a kind of thing no video player has. For most groups of students in this class there is no library to meet at instead, no dorm lounge, no lab bench; the alternative to meeting in the virtual hospital is, as one student noted, social media platforms like Discord or not meeting at all.⁵⁰ If the world’s difference from the video of itself is to be made of anything, it must be made of the meeting.

Rhetorical incorporation is one in substance and dual in being. A world becomes a student’s own in two ways at once: as a place they dwell in—world-disclosedness, the register every presence finding so far has measured—and as a place that is theirs together with others—co-disclosedness, the same world had as ours. The two are not a feature and its supplement; they are two aspects of one phenomenon, and the chain itself grounds the second. Desire’s thymotic species is irreducibly social: one is angry at someone, stands on honour before someone; the emotions that concretize judgment into act are articulated toward or before determinate others—anger toward the slighter, shame before the audience that matters. An act, when it finally comes, is rarely aimed at no one; the chain’s last link carries a toward-whom as part of its structure. What Burke names consubstantiation—persons made, through acting together, of one substance—names the same ground from rhetoric’s side: to act with another is to be with them in a way no shared spectacle alone provides. A world that supports action will support it toward and with others, or it will support less than action’s full structure.

The three routes hold this dual structure very differently. The published headset version maximized world-disclosedness: the encircling display, the fullest supply of the operating room a medium could give. Co-disclosedness it did not carry. Students entered alone, and we said so in print, naming among the version’s limits “the sheer lack of features found in traditional games like multiplayer functionality, voice chat, and

⁴⁹Course structure and graded instruments per the course syllabus, BME 179: Biomedical Engineering Design: Addressing Unmet Clinical Needs (University of California, Irvine); the same syllabus has been used across the offerings surveyed.

⁵⁰“The virtual space to meet up for teamwork was very smart, but my group utilized discord a lot” (S25-R072/Q10). The course itself maintained a Discord channel for the whole class in every term surveyed, so the nearest alternative meeting place carried institutional standing of its own.

customizable avatars” (“Phenomenological Evaluation” 390). The YouTube platform possesses neither in kind: it is a player, not a world; there is nothing in it to be in, together or otherwise. The desktop route inverts the headset’s profile. Its world-disclosedness rests on the thinnest supply: the monitor gives less, and *phantasia* furnishes more of the world. But as the program stands deployed, it is the only route on which co-disclosedness is available at all. The scope of that claim is historical, not metaphysical: proximity voice chat could be built into a headset version—nothing in the hardware forbids it—but development moved to the desktop build, and the social possibilities were built only there. The affordance is not a consolation native to screens; it is a developed asset this platform alone carries.

What the meeting changes is not the platform’s feature list but the being of the world. A room entered alone is a viewing station; the same room entered with another student is ours before anything is said in it—a place jointly attended, where the judgment forming over the footage can be spoken, answered, and made common. The research on learning at a distance has a name for this register of experience and a hard-won result about it. Social presence—defined at its core as “the sense of being with another” (Biocca et al. 456)—is not a property a medium contains but a state persons achieve, something a course can activate or let lie (Terry and Doolittle 121). Those findings were made about text forums and video calls, not about rendered worlds, but the insight is applicable all the same: co-disclosedness, being-with-others is a second presence, of its own kind, that the desktop hospital can host—and the first kind, presence in a place, cannot substitute for it.

A world, though, can only hold a meeting open; it cannot convene one. Footage is delivered to whoever presses play, and a place is there for whoever walks in, but being-with requires that others arrive. Arrival is exactly what the deployment’s shape made unlikely. Convening a group asks what solo watching never asks: schedules aligned across campuses that share no calendar, an invitation risked on classmates never met, proactive organization, and commitment to actually meet with each other. What that commitment can come to in practice, one student recounts in full: “They reached out to me in order to begin the group project, and I responded, and I was ghosted until the final few weeks of the quarter where I found myself very behind in the group’s work with no knowledge that they were organizing without me” (W26-R036/Q6). The account concerns the group’s work generally rather than the virtual hospital, but it shows the social ground the meeting had to stand on: an invitation extended, answered, and dropped. The collaborative-learning literature warns of precisely this: interaction cannot be assumed merely because an environment makes it possible (Kreijns et al. 336). The collected data shows the warning borne out. In 873 responses there is one confirmed convening—the student who wrote that “you can watch with others” (W26-R037/Q10). Eleven students wrote of being alone in the world; “when I did log in, I was usually alone in the platform,” one of them explained, “so it didn’t really make me feel more connected with my peers” (W25-R038/Q10). Four asked the course to require the meeting; thirteen asked for the capability, twelve of them in wording implying it did not exist. The affordance that answers the world’s justification-question was unknown to part of the class that wanted it, unconvened by nearly all of it, and lived by one voice in three terms. The world held the WE open; the course’s temporal shape and lack of designated tasks let it lapse.

The deployment’s deepest loss now has its exact name. It is not presence—presence the desktop world had, measured and attested. It is the WE left unconvened: a co-disclosedness the ecology held open for three terms and was predominantly unfulfilled. The arc that began this analysis closes on itself. The difference we announced at the course’s outset was the meeting; the audit of materials arrived at the meeting; the survey’s most skeptical voice, unprompted, reinvented the meeting. The announcement was right, and what it named went largely unlive. The question the presence analysis left open—what could bind a world to a student exclusively, when its footage plays anywhere—has the same answer: its others. The virtual hospital is irreplaceable exactly insofar as it is shared, and replaceable exactly insofar as it is not. Whether a convened WE would also blunt the boredom this analysis diagnosed—company giving sustained interest a completion of another kind—the collected data cannot say; the suggestion is speculative, and it is offered as such. What the data can say is narrower and firmer: the one thing this world alone could be was the one thing it was almost never used as.

Two findings from this deployment reach beyond it. The first is the pair of matched numbers: the pilot’s 70% under headsets, the deployment’s 70.7% before ordinary monitors. The supply could hardly differ more—an encircling display against a screen—yet the felt result barely moved. Parity of that kind is what the leveling power of *phantasia* looks like in data: where the medium supplies less, *phantasia* furnishes more, and the felt world arrives nearly whole either way. The second finding is what no route could level: the

act that none of them contained, and the meeting that only one of them carried. Route sets the load; the task sets the dwelling. The first half of the motto names the chain’s opening stations: the route—headset, desktop, browser—decides how much of the world reaches the student through the senses and how much *phantasia* must furnish from within (A_1 into A_2), and the matched numbers show *phantasia* equal to any load a route set it. The second half names the chain’s end. Dwelling is world-disclosedness sustained over time: not the momentary feeling of being placed in a world, but the settled being-in of a student who returns to the world, moves through it toward something, and has their doings answered by it. What converts a place merely felt into a place dwelt in is the act—the passage from committed judgment (A_3) into an actualized potentiality (A_4). That passage occurred on every route, because watching is itself the actualization of a potentiality the world furnishes; the deployment’s students actualized it by the hour. The design question is therefore never whether students will actualize a potentiality—they always will—but which potentialities the world makes available, which one each student actualizes, and why that one rather than a richer one. At the scale of 241 students, the deployment attests that this is where worlds are won or lost: at the passage from A_3 to A_4 , in the worth of the potentialities a world furnishes to the commitments it has solicited.

6 Design: The Levers as *Archai*

What remains is design, and the diagnosis dictates its form. A design lever, in this analysis, is an *archē*: Aristotle’s fourth sense of the word names “[t]hat from which ... a thing first arises, and from which the movement or the change naturally first proceeds” (*Metaphysics* V.1, 1013a7–10)—the productive source from which a change first comes. To call something a design lever is to make a two-part claim: that one of the deployment’s documented failures—the quarter of the class stopped at the installer, the forty-seven students who chose YouTube over the world, the group meetings that never convened—has a definite productive source in the design itself, and that design can reach that source and change it. The five levers that follow are ranked by the station of the actualization chain each one works on: the way into the world (A_0), the act the world houses (A_3 to A_4), the object the seeking is for, the convening of the WE, and the shape of the watching’s time. Ranked this way, the levers recapitulate the diagnosis: each names a place where the deployment fell silent, and the productive source design must work on there.

Five levers carry that claim, and each can be given a plain name: the *door*, the *task*, the *find*, the *meeting*, and the *while*. The *door* is the way into the world—the download, the installation, the sign-in, the machine all of it must run on. A quarter of the class never got past it; for those students the world disclosed nothing, because it never opened. Everything design does inside a world waits on this lever, since a world is only as good as its way in: a browser-deliverable build, Mac support, and entry without setup re-admit the students the *door* turned away. The students proposed the same remedy themselves: “Provide a cloud version of the VR. Meaning, provide the VR setup and ready to go on a cloud machine provided by UCI. This will help for inclusivity in the VR” (S25-R059/Q9)—entry without local installation, asked for in the name of exactly the students the *door* excluded.

The *task* is what there is to do once inside, and its absence is the obstacle the deployment never solved. As deployed, every assignment the course gave could be completed without entering the world; the watching was mandated, but nothing had to be done where the watching happened—and students asked for exactly what was missing.⁵¹ The field’s founding account of what three-dimensional learning environments are for named this obstacle in advance. What such environments distinctively offer, Dalgarno and Lee argue, are tasks: “well-designed 3-D VLEs can enable learning tasks that are not possible or not as effective in 2-D environments” (“Learning Affordances” 25)—which is what the student’s surgical platform proposes, almost to the letter. The same paper issues the warning this deployment went on to demonstrate: comparisons that hold the learning design constant across a 2-D and a 3-D environment “would be likely to only demonstrate the trivial fact that if the unique affordances of 3-D VLEs are not harnessed within the learning design, there will be minimal unique learning benefits” (25). The deployment ran that comparison without meaning to—the same design, watch the footage, on both routes—and the cohort returned the predicted verdict in its own words: “the youtube videos were enough” (W26-R033/Q9). A second proposal closes a loop the analysis left open. The student whose praise stood over an emptiness they could not name—the divergence

⁵¹One proposed it in detail: “Maybe an interactive surgical platform where a group can work together to dissect/do surgery to further understand what a surgery consists of ... it has steps so a group can virtually perform the surgery” (W25-R035/Q10).

register’s culprit-less voice—returned, in another answer, naming the missing thing exactly: “I think there should be reflections in the VR environment. While being immersed in the procedure is fascinating, having reflections afterwards would allow us to look for the main takeaways from these procedures and allow us to think critically about clinical needs” (W26-R016/Q9). What the second-form echo could not locate, the same student’s design request supplies: an act of reflection, housed where the watching happens. One way to remove the obstacle is requirement: make the work possible only inside the world, and the rooms will fill. A compulsory world is still a world. What compulsion changes is how the world is entered and felt, and Aristotle explains why: what is done under compulsion runs contrary to nature, and for that reason it is painful—felt as force unless custom has long since softened it.⁵² The dissertation’s earlier analysis of a tutorial world, *Gnomes & Goblins*, shows the alternative in miniature. That experience also needed a beginning—no world starts without one—and it designed its start button as a first lesson: to enter, the player must light a candle, and the fumbling the lighting process takes introduces the grip, the trigger, and the world’s whole philosophy of exploration over instruction. The requirement did not disappear; it was made to teach. The deployment’s version of the same design is choice. Let the assignments remain completable outside—the YouTube platform and the course page stay open—and build the environment so that everything can also be done in one place, inside the world, where the footage, the finds, the notes, and the group already live. A student who enters on those terms enters by their own act. The force Aristotle warns of drops out of the equation—or, where the choice is more arranged than open, out of conscious recognition, which for the experience amounts to the same—and the world gains a convenience no other route offers: the course’s whole work gathered under one roof.

What the *task* should be built toward, the actualization chain specifies link by link. The chain traces how a world becomes an act: perception takes the world in; *phantasia* renders what was taken into a determinate image; judgment commits to what the image presents, and emotion concretizes in that commitment; desire, so formed, completes itself in an act. A task built on the chain gives every link its material inside the world: footage worth the taking-in; prompts that give the forming image something determinate to resolve on; judgments the deliverable asks the student to commit to and defend; and an act—the need noted, captured, submitted—housed in the same rooms where the seeing happened, so that nothing in the world’s workings sends the student elsewhere to complete what the world began. Design cannot manufacture rhetorical incorporation. Rhetorical incorporation is a mode of *being-in-the-virtual-world*: it is something curated, a harmonization between the student and the virtual world within which they dwell; what design does is make it achievable, clearing the way at every link. Build the task so, and the world’s existence stops needing an argument: in every way, shape, and form that matters to the student’s own end, the world is perceived as better than any alternative.

The task lever’s pedagogical warrant is already established at meta-analytic scale. Across health-professions education, training that puts the learner’s hands on the procedure—simulated, not real—has been tested against no training at all in 609 studies enrolling 35,226 trainees, with large pooled effects on knowledge and skills; the evidence is so uniform that the reviewers close by doubting the comparison is worth running again: “with only 4% of the outcomes failing to show a benefit, we question the need for further studies comparing simulation with no intervention” (Cook et al. 987). That meta-analysis is medium-agnostic—mannequins and cadavers count alongside screens—so it grounds the task’s warrant without ranking routes; the finer work belongs to the closest analogue this deployment has in the literature. Dubovi, Levy, and Dagan built the kind of world the student’s proposal asks for: a non-immersive desktop simulation of a clinical procedure—medication administration, in a virtual hospital ward—where nursing students assess the patient, prepare the dose, administer it, and monitor what follows. Students who trained in that world scored higher on the study’s knowledge measures than a matched cohort taught the same material by lecture, and the advantage was still undiminished when the students were retested five months later (23). The study’s sharpest finding is the watching-versus-doing distinction made empirical: of ten students whose sessions were recorded, nine answered the procedure worksheet correctly, and nine nonetheless selected the wrong medication on their first attempt inside the world—errors that fell to near zero by the second scenario (23–24). Knowing the procedure on paper and being able to perform it are different attainments, and only a world with a task in it exposes the difference and then closes it. The authors’ design principle is the task lever’s own conclusion, reached by measurement: “ensuring that participants gain sufficient control

⁵²“So all acts of concentration, strong effort, and strain are necessarily painful; they all involve compulsion and force, unless we are accustomed to them, in which case it is custom that makes them pleasant” (*Rhetorica* I.11, 1370a12–14).

of the environment is an important condition for learning,” for “presence is not a unique attribute of the environment, but it is induced by the fidelity of representation and the high degree of interaction and user control” (25). That last sentence reaches past the task to the door and the rooms: nothing about a desktop world forecloses presence; fidelity—to a lesser extent—and interactive control and agency—to a much greater extent—produce it.

The *find* is the object of desire, and here the diagnosis must be exact, because the student under mandate is not desireless. Every student arrives carrying a governing desire that organizes the rest: to pass the course, for the sake of the degree, for the sake of the life the degree serves. In the vocabulary this dissertation has carried from Heidegger, that is the student’s *for-the-sake-of-which*—the end from which every decision, conscious or otherwise, is perceived, evaluated, understood, and enacted. What varies beneath that end is the way it is pursued: which platform to open, where to watch, whether to watch alone. Watching on the YouTube platform and watching in the virtual hospital are, to that governing desire, two routes to the same end, and if the routes look equally effective, the more convenient one wins every time. Curating the *find* therefore means shaping the world so that the student’s own weighing lands inside it. The first condition is parity: the build must at least equal the YouTube platform in video quality and in everyday conveniences, because no curated object survives a worse player. Past parity come the elements only the world can add. Prompts name, for each procedure, what a needs-finder watches for, so the looking has a target. Captions and transcripts mark the moments that matter, so the target can be found and cited. Each find a student captures should feed directly into the group’s graded deliverable—an unmet need identified in the footage becomes an entry the innovation notebook and the final report are built from—so that the desire for the find is not a new desire the course must create but the student’s existing desire for the grade, extended one step down into the world. The meeting waits inside, worth the *door*’s inconvenience. Each element gives the governing desire one more reason to choose this room. The choice must remain real. The YouTube platform stays open for the quick second viewing, for the student without a computer strong enough to run the environment, and for the one who wants to jump back in quickly to verify something about an operation; the aim is not to close the other route but to make the world the one a free choice keeps landing on. Agency itself becomes the pedagogical instrument: the student moved by being handed a world, never a word.

The *meeting* is co-disclosedness convened. The proximity voice chat affordance exists and every student already belongs to a procedure group; what design must produce is the convening itself—whatever brings two students into the same room at the same hour. It is the one lever whose product no video can imitate, and the *find* can be shaped to invite it. A procedure fills the frame with more than one stream of activity—the surgeon’s hands, the instruments passing, the movements of the team around the table—and no single watcher attends to all of it at once. A prompt that assigns two vantage points, or a find that asks what happened at the table to be reconciled with what happened around it, makes a second pair of eyes worth more than a second viewing: the desirable object doubled into an occasion to meet. The *while* is the temporal frame—the same while that boredom stretches. The length complaints ran through every term, but the cut they imply would be the wrong lever. A procedure edited for pace is no longer the procedure, and what an edit removes may be exactly the unremarkable corner of the operating room where an unmet need sits; the footage must stay whole. The watching does not have to. Segment the procedures into portions assigned week by week, and set interactive sequences at intervals inside the watching—prepared moments that interrupt the drag, catch the attention as it wanders, and redirect it to what matters. The students asked for this articulation in nearly those terms. One tied the length directly to lost focus: “The long videos of the procedures in the VR could probably be improved. I didn’t really understand what was going on, and the videos felt too long. If they were split into multiple parts it would probably be easier to focus on the procedure” (S25-R017/Q6). Another proposed the weekly portioning with its diagnosis attached: “I think a good think would be to assign certain VR on video to watch weekly so that all members could stay on track. because the main issues was people were just waiting too last minute to watch the videos” (S25-R037/Q6)—segmentation not only against the drag of the while but against the procrastination a term of total freedom invites. The while is not shortened; it is articulated. Session length, sequence, and rest belong to the levers, not the logistics.

Of the five levers, the *meeting* asks the most of design, because everything needed for it already existed and it lapsed anyway. Co-watching is a coordination game. Two students who would each get more from watching together can each still watch alone, and the solo path asks nothing of them: no schedules aligned, no invitation risked, no hour kept. The meeting asks all three, and it asks them of all students in the same

group simultaneously. A coordination game of that shape has two stable outcomes—everyone convenes, or no one does—and which outcome a cohort lands on is decided by the default. Left to itself, the default decides for solitude: the solo path’s frictions have long since been engineered away by the platforms that host it, while the meeting’s demands stand untouched. What the collected data showed is therefore an equilibrium, not a verdict on the students: co-watching solved in principle—the affordance built, the groups assigned—and unsolved in practice, one student confirmed convening in the three terms the class has run.⁵³ The lever’s name is exact. What must collapse is the activation energy of the meeting: the demands that stand between an assigned group and its first shared hour.

Every remedy for the meeting draws on one material: shared time. The course was built, deliberately and for good reason, to need none of it. Its students enroll across nine campuses, quarters set against semesters, with no timetable in common; a synchronous version of the course was never in the design space, and an in-person one still less. Whole-cohort synchrony is therefore closed to design. What remains open is small: the overlap of two schedules, a procedure group’s single shared hour—micro-synchrony, the dyad-sized coincidence the systemwide constraint still permits. Time, in this frame, is a design material. The deployment allotted all of it to access—any student, any hour, any place—and that allotment was right; the error was only that nothing was held back. The meeting needs a small reserve of shared time, and the deployment reserved none. The design problem is exactly that narrow: not to restore the classroom hour the course simply could not implement, but to reserve, inside a schedule built on total freedom, the one small overlap the WE requires.

The remedies can be ordered by how much shared time each asks of the group, and the first asks none: demonstrate the capability. Twelve of the thirteen students who asked for a way to watch together asked in wording that implied the platform had none—“if there was a way to interact with other users who are online at the same time” (W25-R071/Q9)—and this despite the course page, which states plainly that groups can meet and talk inside the environment. The capability was described in writing and asked for as if absent; what description did not accomplish, demonstration can—an onboarding walkthrough, a live demonstration in the first week’s module, the capability seen in use rather than read about. The second remedy also asks no shared time: let the world carry traces of the group—who entered today, which rooms a groupmate visited. A trace shows a student that their group uses the space; what it cannot do is bring two students into the space at the same hour. Watching together by proximity voice chat happens only when both students are present at once, and no sign of yesterday’s visit creates today’s meeting; traces point toward the meeting without producing it. The third remedy is an assignment, and it produces the meeting’s hour while leaving the choice of hour entirely with the group. Early in the term, each procedure group uses a group-scheduling tool to settle the weekly time its members can share; the time is the group’s own finding, not the course’s imposition. The students suggested this remedy directly themselves: one proposed “setting up a class google sheet where times are set by each group to enter the VR program/watch the videos” (W25-R060/Q10); another, a “scheduled online mixer or networking session early in the course” to make finding groupmates easier (S25-R079/Q6). The second proposal reaches past scheduling into affiliation, and the collaborative-learning literature says it must: Kreijns, Kirschner, and Jochems name as their second pitfall “the tendency to restrict social interaction to educational interventions aimed at cognitive processes” (336)—a meeting made only of assignments starves the social ground the meeting stands on. The students also supplied a reason the world’s own meeting-form can carry that social ground better than a video call: “I think a virtual group will be nice, students could make their own characters, students who don’t know each other may shy to speak in reality, but if there is a virtual image for them, they might have less stress talking to others” (W26-R001/Q10). Two more voices corroborate the mechanism from the camera’s side, asking for “group communications that do not wish to do video camera meetings” (S25-R002/Q10) and noting that “some of us don’t wanna have their cameras on” (S25-R074/Q10). The avatar lowers the exposure the invitation risks; for strangers assigned to one another across nine campuses, that reduction works on exactly the activation energy this lever must collapse.

The fourth remedy closes the design with accountability, still without imposing a timeslot. Each group’s weekly meeting carries a brief follow-up assignment: minutes, submitted after the session—what the group

⁵³It is likely that more than one student convened in the space with their group. The survey, however, was built without the foresight of a question that asked directly whether the assigned group had used the environment to meet, so the collected data cannot say how many groups actually did. The count is a floor—the number of responses that happen to confirm a convening—never a measure of how many occurred.

watched, what it found, what it settled. The occurrence is graded; the timeslot never is; the hour remains the group’s own settlement from the scheduling assignment. Small meetings are the right form. In immersed collaborative environments, learning gains run inverse to group size—smaller groups, larger gains (De Back et al. 5375–76)—and reviews of monitor-based collaborative worlds judge voice communication essential (van der Meer et al. 9–11); groupmates chatting through proximity voice chat—two students, one procedure, one shared hour—or the full procedure group at once, is exactly the form the evidence favors. A requirement stands at the start of this design, and Aristotle’s qualification, quoted above, explains why it can stand there without poisoning what follows: the forced is painful “unless we are accustomed” to it, and then “it is custom that makes” it pleasant. The assignment’s work is to establish and naturalize the custom—to bring each group through its first meetings, while the demands are still unfamiliar and the worth still unexperienced. What it curates is, in Aristotle’s terms, a habit: among the seven causes of action, the one that stands nearest to nature, for “as soon as a thing has become habitual, it is virtually natural” (*Rhetorica* I.11, 1370a6–8). If that were all the design ever did—if, term after term, students met only because minutes were due—the meeting would have remained what the seven-causes analysis names compulsion: an act whose origin stays outside the agent, done under force and felt as force, yielding compliance rather than practice. The design intends the opposite sequence. The requirement produces the first meeting; the first meeting lets the students experience what the survey’s one confirmed convener reported—“[t]he main pro to the platform is you can watch with others” (W26-R037/Q10)—and from then on the group meets because it wants to, the cause now inside the students rather than in the requirement. The goal, throughout, is exactly that movement: an available affordance turned into a consciously selected practice. That movement is also a discovery the actualization chain makes diagnosable. Requirement and practice can look identical from the outside—the same group, the same room, the same hour—and the chain names where they differ: in the *archē* of the act, its origin lying in the requirement or in the students’ own desire. A course that watches only attendance cannot tell compliance from practice; a course that asks where the act’s origin now lies can see, term by term, whether the affordance is still being forced or has begun to be chosen.

The five levers share one motto: design against the question “why not just YouTube?” Every element of the world must do something video alone cannot—give a task, hold a place, convene a meeting—or the element loses, rationally, to the platform that plays the same footage with less friction. A craft this effective deserves candor about what it is, because the attention economy practices the same craft in mirror image, and the actualization chain shows exactly where the mirror sits. Gloria Mark’s account of attention capture is precise about the mechanism. The algorithms learn “who you are, how you feel, what you do when and where” and use it “to capture your attention” (Mark 155). Their instruments are targeted solicitations built to elicit what Mark calls lower-order emotions—reactions that “occur spontaneously and automatically without cognitive processing”—and they are timed for the moments “when you are low in attentional resources or in a bored or a rote state,” when “you can’t help but respond to them” (159–60). Read through the chain, the design is exact. A bored viewer still desires—the governing end never sleeps—but the moment offers no determinate object to want. The application supplies one, pre-charged: perception is seized by the notification; the image arrives already pleasurable; and the act completes before the cognitive orientations have done their work—no deliberative weighing of the object, no committal judgment about its worth. Nothing in the chain is broken by such design; the chain runs. The pedagogical craft uses the same knowledge in the opposite direction: it curates a worthy object of desire, informs the judgment that weighs it, and gives committed desire an act to complete in the world—every link furnished, none bypassed, the completion serving the student’s own *for-the-sake-of-which* rather than a platform’s. Wendt states the designer’s side of the same boundary: what designers make acts on the world long after they are done with it, the nefarious end of the spectrum runs from persuasive design to dark patterns, and the calling that stands against it is “advocating for preferable worlds” (*Design for Dasein* 162). To design this way is to hack the soul. That is no reason to refuse the craft; it is the reason the craft bears the responsibility it does, and must answer for the dispositions it lays down—the settled ways of looking, judging, and meeting that outlast the course. A world that teaches by being dwelt in shapes its dwellers, and the designer’s obligation is to shape them toward their own good—openly aimed, and defensible to the very students it moves.

7 Limitations

Five ceilings bound everything this section has claimed, and each has a consequence worth stating plainly. The survey is a single channel of evidence, so the second form of boredom appears in this analysis only as echo, never as detection. Wherever a response was read as engaged on its surface and empty underneath—a student conceding the world’s immersion while reporting that nothing in it was the highlight—the surface came from the student’s words, but the emptiness came from Heidegger’s frame; no second measure of these students exists to confirm or overturn such a reading, so every claim of that shape stands or falls with the frame. The instrument is retrospective, so felt time is nearly mute in the collected data; where the analysis has spoken of the while stretching, it has spoken from the form’s grammar and from the laboratory’s separate evidence, not from these students’ reports. The survey’s three content items—the questions testing whether students grasped the procedures themselves—drew correct answers from nearly every student in every cohort; scores bunched at the top of a scale leave no variation to analyze, so those items cannot show whether one format taught better than another, and no claim in this section concerns learning outcomes. Effectiveness, wherever the word has appeared, is scoped to what the chain-completion evidence can carry—presence attested, interest sustained, acts housed or unhoused—and the design section’s “better than any alternative” names an aim whose test would be the redesigned deployment itself, not a result already in hand. The three cohorts differ in size and likely in composition, so the arcs—frictions falling, the value-question rising—are patterns observed but not yet definitive. Heidegger’s account of boredom, finally, supplied this section with categories, never with findings: the three forms, the paired moments of being left empty and being held in limbo, and the two-channel rule functioned as a scheme for sorting what students wrote—which complaints name a culprit and which do not—rather than as instruments that measured an inner state. A fundamental attunement (*Grundstimmung*) is, on Heidegger’s own account, something no retrospective questionnaire answer could establish; no reading in this section claims to have detected one, and none should be read as if it had. What survives these ceilings is the section’s actual yield: where the world fell silent, what its difference from the video of itself must be made of, and how design can convene what the collected data shows was wanted.

8 Implications and Close

The virtual-learning literature already measures something it calls boredom, and the shape of its instrument shows exactly what is missing. In the field’s most instrumented media-comparison experiments, boredom is the name of the lowest band of an EEG workload metric: whatever falls beneath the scale’s lower threshold is classified as “boredom,” so that the word means nothing more than workload’s absence—the bottom of a single axis (Makransky, Terkildsen, and Mayer 229–30; Parong and Mayer, “Cognitive and Affective Processes” 234–36). One of those experiments shows why the label is not enough. In Parong and Mayer’s media comparison, the students in the headset reported the highest presence and the highest enjoyment in the study, with large effects on both; the same students were classified into the boredom band at a significantly higher rate than the slideshow group, and they learned less by the study’s transfer measure (“Cognitive and Affective Processes” 233–36). The self-report channel found those students full; the physiological channel found them empty; the outcome measure sided with the empty reading. That is the divergence this section could only hear as an echo in a student’s prose, produced there by independent instruments in a single published experiment—and the study can read it only as misallocated workload, because boredom-as-low-workload has no structure: it cannot distinguish the student searching for something to attend to from the student occupied but held by nothing from the student withdrawn from the world altogether. Heidegger’s account distinguishes exactly these, and no published study joins it to an instrumented reading of learners inside a virtual world. The combination is unbuilt, and the laboratory study this Part turns to next was designed inside the frame that makes it buildable: a surface reading of what occupies the student run concurrently with a depth reading of what holds them, the disagreement between the channels serving as the diagnostic.

The unconvened WE leaves the empirical program a narrower question, and it comes with its own apparatus attached: what is the least shared time that convenes the WE? The design section’s four remedies already stand in the order the question requires, by how much shared time each asks of a group—the demonstration asks none, the traces ask none, the scheduling assignment asks one hour of the group’s own

choosing, the minutes make that hour accountable—so the redesigned deployment is itself the instrument: introduce the remedies in that order, term by term, and watch for the increment at which students begin to meet. The collected data has already fixed the baseline any such observation starts from—thirteen students asking for the meeting, one confirmed convening—and the question’s answer would be a finding about shared time as a design material: how small a reserve of it suffices. A further question makes itself known for the first time.

The desktop clinic completes a test the dissertation’s frame has been running since Part I. Part I built the actualization chain from perception to act. Part II built rhetorical incorporation—one in substance, dual in being as world-disclosedness and co-disclosedness—together with the demand of veri(dis)similitude, and demonstrated the frame against the richest of authored worlds: story-driven, hand-crafted, entered by choice. This section has now tested it against the sparest world the dissertation examines: a virtual hospital of plain rooms and procedure footage, equipped with proximity voice chat, deployed under course mandate to 241 students. The vocabulary held. Everything those students met was authored—the rooms, the footage, the hail spoken before the world first rendered—and none of that authorship determined an outcome. The chain ran always; each student’s acts completed where that student’s own judgment routed them, most often in the alternative the course itself had furnished. What the frame contributes is the ability to say precisely what stood in the gap between total authorship and undetermined outcome. Without it, the survey’s answers read as a feature backlog and a usage shortfall. Within it, they resolve into a three-part diagnosis: a world that fell silent at the chain’s final link, since the act the course assigned—analysis of the procedure—could be completed entirely outside the world, and inside it nothing beyond watching was ever asked; a flight that was never a malfunction, since the chain merely shows each student following their greatest desire—both routes served the governing end, and the greater desire lay with the more convenient route; and a hail whose one unmatchable promise, the meeting, went unconvened for want of a small reserve of shared time. What the analysis demonstrates, at the scale of 241 students, is the claim the section opened with. A virtual world on a flat screen can place its students: seven of every ten application users attested presence on an ordinary monitor—the pilot’s headset rate, with no headset. Presence alone does not justify a world: every video in the virtual hospital also played one click away, students weighed the world against the video of itself, and wherever the world furnished nothing beyond watching, they completed the chain through the more convenient route. What could have justified the world is the one capability the copy cannot reproduce: watching together, in the only room this class has ever had. The world held that capability open for three terms; one voice in 873 responses confirms it was ever lived. A wholly authored world, in short, attunes its students without determining them—and this deployment has demonstrated both halves of that sentence at cohort scale. That is what the dissertation carries forward from this section: not a premise defended, but a finding held.

Appendix A: The Published Pilot Study

[The pilot article — King and Salvo, “Phenomenological Evaluation of an Undergraduate Clinical Needs Finding Skills Through a Virtual Reality Clinical Immersion Platform,” Biomedical Engineering Education, vol. 4, 2024, pp. 381–97 — is reproduced here in the final dissertation assembly.]

Appendix: Catalogue of Reported Frictions

Every friction students reported is collected here, ordered from most to least reported. A complaint is counted once per coded application under the friction family of the codebook; the percentage columns count students, not complaints—the share of the 123 application users, and of the 108 students on the YouTube route, who raised the friction at least once. Two of the frictions we predicted during development appear at the bottom of the table precisely because the survey did not bear them out.

Friction	Complaints	App users / YouTube	Example from the survey
Crashes and freezes	29	15% / 8%	“I had issues with crashes” (W25-R064/Q6)
Install and compatibility	26	3% / 17%	“[m]ore inclusive for macbook users :”(W25-R042/Q9)
Buffering and latency	25	11% / 8%	“maybe improving the lags, I would glitch sometimes” (W25-R084/Q9)
Wayfinding and navigation	21	11% / 6%	“eliminate having to go up stairs and stuff?” (W25-R084/Q9)
Hardware access	13	3% / 7%	“the file being too large, my computer could not handle it” (W25-R062/Q9)
Video fidelity	10	5% / 3%	“I feel like it’s very blurry in my experience” (W25-R062/Q9)
Viewing proximity	2	—	“surgical screen far in the back which made it hard to see” (S25-R038/Q10)
Avatar occlusion	0	—	(predicted; unattested—most students entered alone)

Table 1: Reported frictions by frequency. Counts are coded applications; percentages are students mentioning the friction at least once.

A further 32 complaints name frictions outside these categories and are carried in the codebook as a residual group. Grouped descriptively, they comprise: difficulty understanding the footage or the task it served (5, e.g. “hard to tell what is happening in them,” W25-R067/Q6); interface usability (5, e.g. “The VR platform is not very user friendly,” W26-R039/Q9); distraction and overstimulation (4, e.g. “more distracting than anything,” W25-R092/Q9); motion discomfort and disorientation (4, e.g. “I get motion sickness when watching the VR videos,” W25-R054/Q9); residual crash, freeze, and loading complaints phrased outside the crash category (4); missing or inaudible audio on the surgical footage (3, e.g. “Most surgery videos are silent,” W25-R039/Q9); passivity and filler time (2, e.g. “a lot of filler space/period of inactivity,” W25-R025/Q9); and five single mentions: wayfinding signage, account security, solitude in the platform (“I was usually alone in the platform,” W25-R038/Q10), setup abandonment, and general doubt of benefit. Not counted as frictions are the 19 students on the YouTube route (18% of that group) who simply reported never using the platform at all.

Appendix: Survey-Coding Methods

This appendix documents the coding instrument summarized in the methods discussion: the codebook’s family structure, the reliability results in full, and the merge rules. The complete codebook, all pass files, cross-model comparison reports, and adjudication records are retained in the project archive.

A. Codebook families

The codebook was fixed before coding began and required no structural revision through coding. Counts are total applications in the final corpus of 873 responses.

Family	Scope	Applications	Principal codes
A	Boredom-form signatures	194	determinate-culprit frictions (9 sub-tags), diffuse dissatisfaction, monotony, divergence flags, withdrawal
B	Flight patterns	50	video fallback, minimal compliance
C	Temporal texture	14	time-drag, time-vanishing, objective length
D	Design-node codes (evaluation chain)	256	perceptual conditions, image quality, interest/meaning, action affordances, return-use
E	Involvement channels	259	kinesthetic, spatial, shared, narrative, affective, ludic
F	Incorporation-positive signatures	30	presence, world-language, carry-over
G	Suggestion taxonomy	399	provision, optimization, content, interactivity, guidance, use-cases
H	Emergent (open)	38	coder-invented, dispositioned post-run
X	Housekeeping	269	non-use, contentless, off-topic

Two structural rules governed application. Multi-coding was permitted across families, with best fit chosen per codable idea within a family; and codes name signatures in student talk, never certified inner states.

B. Reliability

Reliability was estimated on a 143-response subsample (16.4% of the corpus), drawn from the term the codebook had least seen and coded by two independent blind passes. Mean Jaccard agreement over full code sets was .878; Krippendorff’s α with MASI set-distance over the same pairs was .82. Per-code agreement (both passes / one pass only):

Code	Both	Pass 1 only	Pass 2 only	Agreement
G6-USECASE	29	0	0	100%
D2-A3	30	0	12	71%
X-OFFTOPIC	25	2	0	93%
X-EMPTY	15	1	0	94%
E-SHA	11	1	2	79%
G3-CONTENT	9	0	0	100%
E-NAR	8	6	1	53%
A1.other	7	1	1	78%
E-LUD	6	2	0	75%
G4-INTERACT	6	0	2	75%
D3-A3A4	6	1	0	86%
A1.lag	6	0	0	100%
B1-FALLBACK	6	0	0	100%
G2-OPTIMIZE	5	0	0	100%
G5-GUIDE	4	0	0	100%
F1-PRESENCE	4	0	0	100%
E-AFF	3	1	1	60%
E-SPA	3	1	0	75%
D4-LOOP	3	1	0	75%
A2-DIFFUSE	3	0	0	100%
A1.install	3	0	0	100%
D1-A2	2	1	0	67%
A1.nav	2	0	0	100%
D0-A0A1	2	0	0	100%
C3-LENGTH	2	0	0	100%
A5-WITHDRAWN	1	0	1	50%
A1.crash	1	0	0	100%

Agreement is weakest on the mention-level channel codes, which the merge handles by union, and strongest on the load-bearing families, which the merge handles by intersection: the design concentrates coder disagreement exactly where it is least consequential.

C. Merge rules

Merging the two passes was deterministic under four rules. (1) Mention-level families (design nodes, involvement channels, suggestions, incorporation signatures, housekeeping, emergent codes) merged by union: for a mention, an omission by one coder is likelier than an invention by the other. (2) The analytically load-bearing families (boredom-form, flight, temporal) merged by intersection, with a one-pass-only code otherwise queued for manual adjudication; a keyword-warrant path for auto-acceptance existed but never fired—every load-bearing dispute reached manual ruling. (3) Friction sub-tags: all text-warranted sub-tags kept; the generic sub-tag dropped whenever a specific one survived. (4) Divergence flags kept at either-pass sensitivity, with precision enforced by manual review of the resulting register.

D. Provenance

All coding passes in the final corpus were performed by Anthropic’s Claude Fable 5 model, blind and independent, under fixed written instructions. Three batches initially single-coded by a smaller model were discarded and recoded after a measured cross-model comparison (mean Jaccard .51–.67 against the .878 same-model benchmark); no coding from the smaller model remains. The author adjudicated all seven manual corrections and reviewed the apparatus rule by rule. A dedicated two-pass recount of the shared-involvement evidence followed the same protocol (93% exact single-bin agreement, eight boundary disputes author-adjudicated).

Works Cited

- Aristotle. *The Complete Works of Aristotle: The Revised Oxford Translation*. Edited by Jonathan Barnes, 2 vols., Princeton UP, 1984.
- Barrett, Robin C. A., et al. “Comparing Virtual Reality, Desktop-Based 3D, and 2D Versions of a Category Learning Experiment.” *PLOS ONE*, vol. 17, no. 10, 2022, e0275119.
- Biocca, Frank, Judee K. Burgoon, and Chad Harms. “Toward a More Robust Theory and Measure of Social Presence: Review and Suggested Criteria.” *Presence: Teleoperators and Virtual Environments*, vol. 12, no. 5, 2003, pp. 456–80.
- Boellstorff, Tom. “Toward Anthropologies of the Metaverse.” *American Ethnologist*, 2023, pp. 1–10, <https://doi.org/10.1111/amet.13228>. (***** UNVERIFIED: online version-of-record pagination; confirm final vol./no./pages.)
- . “Virtual Worlds and Futures of Anthropology.” *AnthroNotes*, vol. 30, no. 1, 2009, pp. 1–5. (***** verify page range against the printed issue.)
- Burke, Kenneth. *A Rhetoric of Motives*. 1950. U of California P, 1969.
- Calleja, Gordon. *In-Game: From Immersion to Incorporation*. MIT P, 2011.
- Chalmers, David J. “The Virtual and the Real.” *Disputatio*, vol. 9, no. 46, 2017, pp. 309–52.
- Chow, Meyrick. “Determinants of Presence in 3D Virtual Worlds: A Structural Equation Modelling Analysis.” *Australasian Journal of Educational Technology*, vol. 32, no. 1, 2016, pp. 1–18.
- Cook, David A., et al. “Technology-Enhanced Simulation for Health Professions Education: A Systematic Review and Meta-Analysis.” *JAMA*, vol. 306, no. 9, 2011, pp. 978–88.
- Cossio, Samantha, et al. “Cybersickness and Discomfort from Head-Mounted Displays Delivering Fully Immersive Virtual Reality: A Systematic Review.” *Nurse Education in Practice*, vol. 85, 2025, 104376.
- Dalgarno, Barney, and Mark J. W. Lee. “What Are the Learning Affordances of 3-D Virtual Environments?” *British Journal of Educational Technology*, vol. 41, no. 1, 2010, pp. 10–32.
- . “Exploring the Relationship between Afforded Learning Tasks and Learning Benefits in 3D Virtual Learning Environments.” *Future Challenges, Sustainable Futures: Proceedings ascilite Wellington 2012*, ascilite, 2012, pp. 236–45.
- De Back, Tycho T., Angelica M. Tinga, and Max M. Louwerse. “Learning in Immersed Collaborative Virtual Environments: Design and Implementation.” *Interactive Learning Environments*, vol. 31, no. 8, 2021, pp. 5364–82.
- Dubovi, Ilana, Sharona T. Levy, and Efrat Dagan. “Now I Know How! The Learning Process of Medication Administration among Nursing Students with Non-Immersive Desktop Virtual Reality Simulation.” *Computers & Education*, vol. 113, 2017, pp. 16–27.
- Fowler, Chris. “Virtual Reality and Learning: Where Is the Pedagogy?” *British Journal of Educational Technology*, vol. 46, no. 2, 2015, pp. 412–22.
- Garrison, D. Randy, Terry Anderson, and Walter Archer. “Critical Inquiry in a Text-Based Environment: Computer Conferencing in Higher Education.” *The Internet and Higher Education*, vol. 2, no. 2–3, 1999, pp. 87–105.
- Guilford, W. H., M. Kotche, and R. H. Schmedlen. “A Survey of Clinical Immersion Experiences in Biomedical Engineering.” *Biomedical Engineering Education*, vol. 3, 2023, pp. 113–22.
- Gunawardena, Charlotte N., and Frank J. Zittle. “Social Presence as a Predictor of Satisfaction within a Computer-Mediated Conferencing Environment.” *American Journal of Distance Education*, vol. 11, no. 3, 1997, pp. 8–26.

- Heidegger, Martin. *Basic Concepts of Aristotelian Philosophy*. Translated by Robert D. Metcalf and Mark B. Tanzer, Indiana UP, 2009.
- . *Being and Time*. Translated by John Macquarrie and Edward Robinson, Blackwell, 1962.
- . *The Fundamental Concepts of Metaphysics: World, Finitude, Solitude*. Translated by William McNeill and Nicholas Walker, Indiana UP, 1995.
- Heim, Michael. *Virtual Realism*. Oxford UP, 1998.
- King, Christine E., Kadin Diec, and Dalton Salvo. “Scalable Virtual Reality Global Clinical Immersion for Culturally Responsive Engineering Design Skills.” *2025 ASEE Annual Conference & Exposition*, ASEE, 2025, paper 49715.
- King, Christine E., and Dalton Salvo. “Phenomenological Evaluation of an Undergraduate Clinical Needs Finding Skills Through a Virtual Reality Clinical Immersion Platform.” *Biomedical Engineering Education*, vol. 4, 2024, pp. 381–97, <https://doi.org/10.1007/s43683-024-00139-5>.
- Kononowicz, Andrzej A., et al. “Virtual Patient Simulations in Health Professions Education: Systematic Review and Meta-Analysis by the Digital Health Education Collaboration.” *Journal of Medical Internet Research*, vol. 21, no. 7, 2019, e14676.
- Kreijns, Karel, Paul A. Kirschner, and Wim Jochems. “Identifying the Pitfalls for Social Interaction in Computer-Supported Collaborative Learning Environments: A Review of the Research.” *Computers in Human Behavior*, vol. 19, no. 3, 2003, pp. 335–53.
- Lombard, Matthew, and Theresa Ditton. “At the Heart of It All: The Concept of Presence.” *Journal of Computer-Mediated Communication*, vol. 3, no. 2, 1997.
- Makransky, Guido, and Lasse Lilleholt. “A Structural Equation Modeling Investigation of the Emotional Value of Immersive Virtual Reality in Education.” *Educational Technology Research and Development*, vol. 66, no. 5, 2018, pp. 1141–64.
- Makransky, Guido, and Gustav Bøg Petersen. “Investigating the Process of Learning with Desktop Virtual Reality: A Structural Equation Modeling Approach.” *Computers & Education*, vol. 134, 2019, pp. 15–30.
- Makransky, Guido, Richard E. Mayer, Nicola Veitch, Michelle Hood, Karl B. Christensen, and Helle Gadegaard. “Equivalence of Using a Desktop Virtual Reality Science Simulation at Home and in Class.” *PLOS ONE*, vol. 14, no. 4, 2019, e0214944.
- Makransky, Guido, Thomas S. Terkildsen, and Richard E. Mayer. “Adding Immersive Virtual Reality to a Science Lab Simulation Causes More Presence but Less Learning.” *Learning and Instruction*, vol. 60, 2019, pp. 225–36.
- Mark, Gloria. *Attention Span: ***** UNVERIFIED subtitle. ***** UNVERIFIED publisher*, 2023.
- Mayer, Richard E., Guido Makransky, and Jocelyn Parong. “The Promise and Pitfalls of Learning in Immersive Virtual Reality.” *International Journal of Human-Computer Interaction*, vol. 39, no. 11, 2022, pp. 2229–38.
- Merchant, Zahira, et al. “The Learner Characteristics, Features of Desktop 3D Virtual Reality Environments, and College Chemistry Instruction: A Structural Equation Modeling Analysis.” *Computers & Education*, vol. 59, no. 2, 2012, pp. 551–68.
- Merleau-Ponty, Maurice. *Phenomenology of Perception*. 1945. Translated by Donald A. Landes, Routledge, 2014.
- Mikropoulos, T. A., and A. Natsis. “Educational Virtual Environments: A Ten-Year Review of Empirical Research (1999–2009).” *Computers & Education*, vol. 56, 2011, pp. 769–80.
- Moravec, K. R., E. L. Lothamer, A. Hoene, P. M. Wagoner, D. J. Beckman, and C. J. Goergen. “Clinical Immersion of Undergraduate Biomedical Engineering Students: Best Practices for Short-Term

- Programs.” *Biomedical Engineering Education*, vol. 3, 2023, pp. 217–23.
- Parong, Jocelyn, and Richard E. Mayer. “Cognitive and Affective Processes for Learning Science in Immersive Virtual Reality.” *Journal of Computer Assisted Learning*, vol. 37, no. 1, 2021, pp. 226–41.
- . “Learning Science in Immersive Virtual Reality.” *Journal of Educational Psychology*, vol. 110, no. 6, 2018, pp. 785–97.
- Radianti, Jaziar, et al. “A Systematic Review of Immersive Virtual Reality Applications for Higher Education: Design Elements, Lessons Learned, and Research Agenda.” *Computers & Education*, vol. 147, 2020, 103778.
- Richardson, Jennifer C., and Karen Swan. “Examining Social Presence in Online Courses in Relation to Students’ Perceived Learning and Satisfaction.” *Journal of Asynchronous Learning Networks*, vol. 7, no. 1, 2003, pp. 68–88.
- Richardson, Jennifer C., Yukiko Maeda, Jing Lv, and Secil Caskurlu. “Social Presence in Relation to Students’ Satisfaction and Learning in the Online Environment: A Meta-Analysis.” *Computers in Human Behavior*, vol. 71, 2017, pp. 402–17.
- Short, John, Ederyn Williams, and Bruce Christie. *The Social Psychology of Telecommunications*. Wiley, 1976.
- Terry, Krista P., and Peter E. Doolittle. “Re-examining Social Presence: Implications for Digital Pedagogies.” *Technology, Instruction, Cognition and Learning*, vol. 11, 2019, pp. 121–39.
- Tu, Chih-Hsiung. “The Measurement of Social Presence in an Online Learning Environment.” *International Journal on E-Learning*, ***** UNVERIFIED volume/issue, 2002, pp. 34–45.
- van der Meer, Nesse, Vivian van der Werf, Willem-Paul Brinkman, and Marcus Specht. “Virtual Reality and Collaborative Learning: A Systematic Literature Review.” *Frontiers in Virtual Reality*, vol. 4, 2023, 1159905.
- Wendt, Thomas. *Design for Dasein: Understanding the Design of Experiences*. Self-published, 2015.
- Witmer, Bob G., and Michael J. Singer. “Measuring Presence in Virtual Environments: A Presence Questionnaire.” *Presence: Teleoperators and Virtual Environments*, vol. 7, no. 3, 1998, pp. 225–40.